

UNIT - I

Linear Algebra

①

Vector space - Subspace - Linear combination & Subsets.

Linearly Dependent & Independent sets - Basis & Dimension.

— x —

Linear Algebra is not a Normal (usual) kind of Engineering mathematics, where we have Differentiation - Partial Differentiation - Integration - Multiple Integral - ODE - PDE - Vector Calculus - Transforms & so-on

usually this kind of mathematics studied by mathematicians not by an Engineer, but now a days its have some vital Applications in Electronics Engineers.

And we ~~call~~ categories Linear Algebra as Pure (or) Abstract mathematics (The word Abstract means, Its a thing only exists in idea & doesn't have a physical existence.

Linear Algebra have a origin from Set language (theory).

Therefore, First we have some idea about set & Binary operation - Groups - Field - Vector space.

— x —

Defn of sets:-

A set is a collection of well defined objects.

Note:-

The word "well defined" have vital role in the defn of sets.

(ie) "well defined" means we have to say (or) conclude that an element is in the set (or) not in the set; we may able to fix either one, not both, (ie) not in a confuse manner.

"It's there (or) not there"
(~~आपका~~ ठीक ठीक है)

For Ex:-

A = Set of all +ve integers ≤ 6

$$= \{1, 2, 3, 4, 5, 6\}$$

$$= \{x \in \mathbb{Z} \mid x \leq 6\}$$

Read it "Set of all" "such that"

Here it is clearly that $-1 \notin A$, $8 \notin A$, $0 \notin A$
but $1 \in A$, $4 \in A$,

we don't have any confusion about the existence or inexistence of an element.

$\therefore$ The set is well defined.

Example:-

B - Set of all old songs.

Here clearly all M.G.R Songs $\in B$

all T.M.S Songs $\in B$

all Sivaji Songs $\in B$

But Can we say all Rajini songs $\in B$

o: we consider the songs from Petla released on Jan 2019, as old, certainly not.

How can we say ~~6 month~~ Equalise 30 years old song with 6 month old song.

On the other hand,

~~Some most of the~~

Rajini

~~Some~~ most of the Rajini songs are old.

So it always confusing, that

all Rajini songs $\in B$ (or) Rajini song $\notin B$.

This confusion lead by, the term "old" is not well defined.

o: B is not a well defined set.

Now

C - set of all old songs prior to 1980.

then clearly,

Whatever the Rajini songs prior to 1980 includes in C.

— x — x —

Algebra of sets:-

$$A \cup B = \{x \in U / x \in A \text{ or } x \in B\}$$

$$A \cap B = \{x \in U / x \in A \text{ and } x \in B\}$$

$$A' = \{x \in U / x \notin A\} = U - A$$

$$\phi = \{\} \text{ Null set.}$$

Notation & Symbols of Number System:-

N - set of all Natural Numbers

$$= \{1, 2, \dots\}$$

W - set of all whole Numbers

$$= \{0, 1, 2, \dots\}$$

Z = set of all Integers (Integral Numbers)

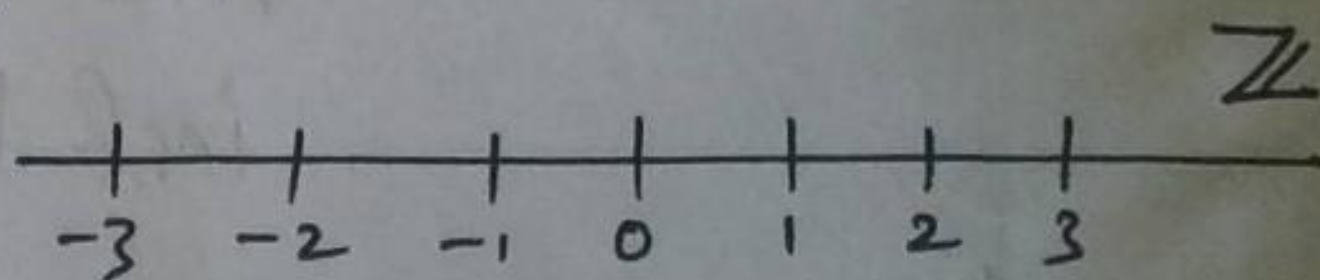
$$= \{\dots, -2, -1, 0, 1, 2, \dots\}$$

Z^+ = set of all +ve integers

$$= \{1, 2, 3, \dots\} = N$$

Z^* = set of all non-negative integers

$$= \{0, 1, 2, \dots\} = W$$



Rational Numbers

Any Number in the form of $\frac{p}{q}$, $q \neq 0$, where p & q are Integers

then $\frac{p}{q}$ is called Rational Number.

Irrational Numbers

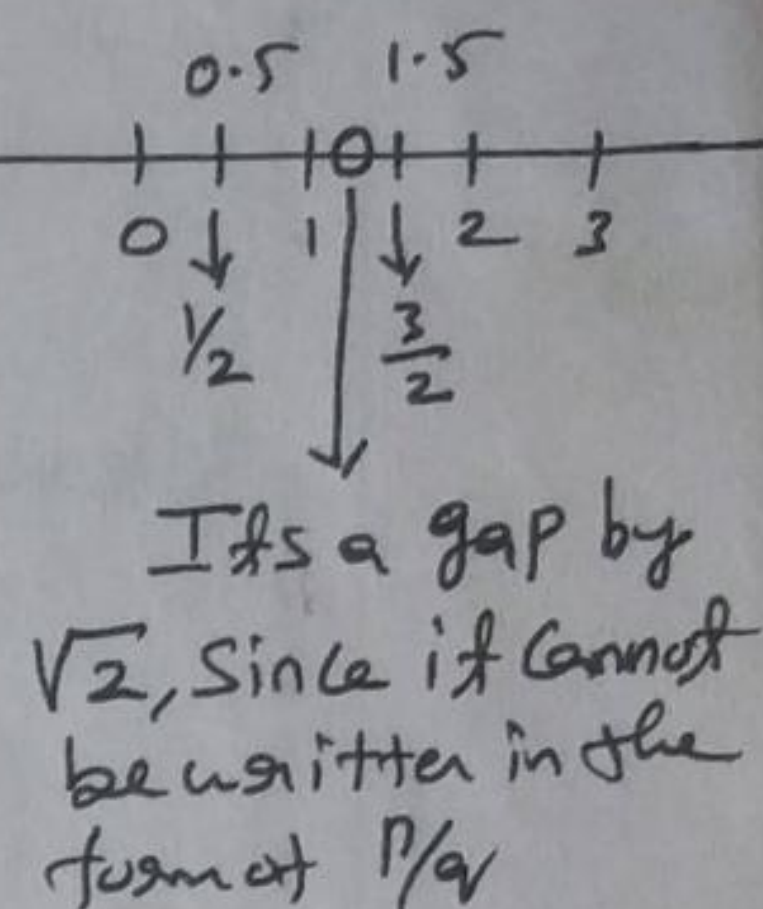
Any Number cannot written in the form of $\frac{p}{q}$, then it is Irrational

$\mathbb{Q}$ - set of all Rational Numbers

$$= \left\{ \frac{p}{q} \mid q \neq 0, p \in \mathbb{Z} \text{ \& } q \in \mathbb{Z} \right\} \quad \mathbb{Q}$$

$\mathbb{Q}^+$ - set of all +ve Rational numbers

$$= \left\{ \frac{p}{q} \mid p, q \in \mathbb{Z}^+ \right\}$$



$\mathbb{Q}^*$ - set of all Irrational numbers

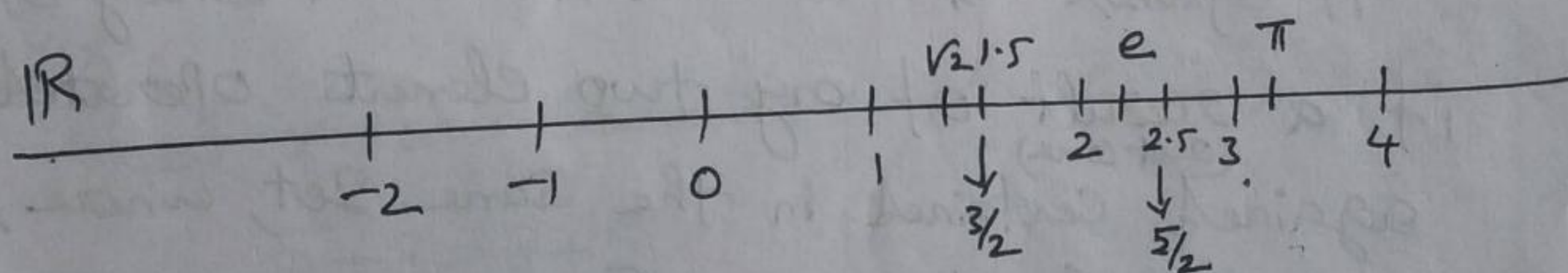
$$= \left\{ \sqrt{2}, \sqrt{3}, \sqrt{5}, \sqrt{6}, \sqrt{7}, \dots, \pi, e, \dots \right\} \quad \left(\pi \approx \frac{22}{7} \text{ not exactly equal} \right)$$

$\mathbb{R}$ - set of all real numbers

Real Numbers

$$\mathbb{R} = \mathbb{Q} \cup \mathbb{Q}^*$$

= It's a combination of both Rational & Irrational numbers

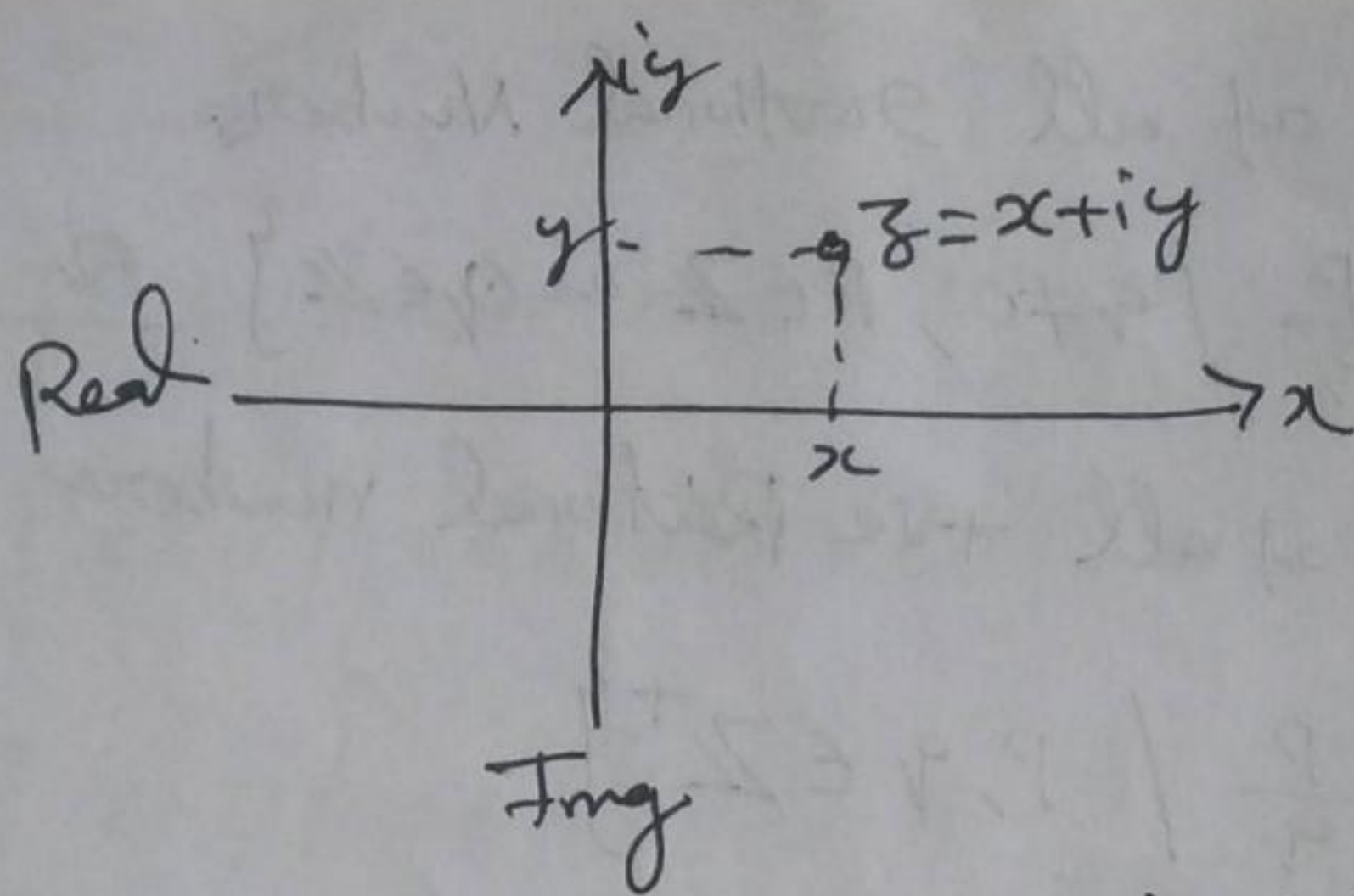


Finally the ^{Infinite} line is completely filled by Numbers called real numbers, so that we call this line as Real line.

— x — x —

Complex Number

$z = a + ib$, where a is real number plot in x -axis
 b is real number plot in y -axis



Eventually, ~~Complex Numbers~~, filled
An Infinite Plane is filled by some numbers
called complex numbers

$\mathbb{C}$ - set of all complex numbers

$$= \{ z \in \mathbb{C} / z = x + iy, x, y \in \mathbb{R} \}$$

_____ x _____ x _____

Binary operation - on a set [closure property of A w.r.t $*$]

A symbol ' $*$ ' is said to be a Binary operation, if a result of any two elements operated by $*$ is again ^(out come) contained in the same set, where, that the two elements comes from.

is $\forall a, b \in A$, then $a * b = c \in A$

then $*$ is binary operation on A

(09)

A is closed w.r.t ' $*$ '

For Ex:-

(4)

'+' and 'x' are (closed) Binary operation w.r.t $\mathbb{Z}, \mathbb{Z}^+, \mathbb{Q}, \mathbb{Q}^+, \mathbb{R}, \mathbb{C}$

But '-' and $\div$ are not Binary Operation w.r.t $\mathbb{Z}^+$ and $\mathbb{Z}$ respectively

(i) $5, 8 \in \mathbb{Z}^+$, but $5-8 = -3 \notin \mathbb{Z}^+$,
 $\therefore \mathbb{Z}^+$ is not closed under '-'

"y" $5, 8 \in \mathbb{Z}$ but $\frac{5}{8}$ (or) $\frac{8}{5} \notin \mathbb{Z}$,
 $\therefore \mathbb{Z}$ is not closed under $\div$

Commutative Property

$\forall a, b \in A$, If $a * b = b * a$, then

then A satisfies Commutative Property w.r.t '*'

(or) '*' satisfies Commutative Property.

For ex:-

'+' and 'x' are Commutes w.r.t $\mathbb{Z}, \mathbb{Z}^+, \mathbb{Q}, \mathbb{Q}^+, \mathbb{R}, \mathbb{C}$

$$5+8=8+5$$

$$(2+i3) \times (4+i5) = (4+i5) \times (2+i3)$$

But '-' and $\div$ are not Commutes w.r.t any set.

$$5-8 \neq 8-5$$

$$\frac{5}{8} \neq \frac{8}{5}$$

Associative Property [Bracket Chasing Property]

$\forall a, b, c \in A$, If $a * (b * c) = (a * b) * c$, then

then A satisfies Associative Property w.r.t '*'

(or) '*' satisfies Associative Property.

For Ex:-

'+' and 'x' are Associative Satisfies Associative Property in any set.

$$(ie) \quad 2 + (3 + 5) = (2 + 3) + 5$$

$$(ie) \quad (2 + i3) \times [(5 + i4) \times (3 + i)] = [(2 + i3) \times (5 + i4)] \times (3 + i)$$

$$(2 + i3) [15 + i5 + 12i - 4] = [10 + 8i + 15i - 12] \times (3 + i)$$

$$(2 + i3) [11 + 17i] = [-2 + 23i] (3 + i)$$

$$22 + 34i + 33i - 51 = -6 - 2i + 69i - 23$$

$$-29 + 67i = -29 + 67i$$

$$\begin{array}{r} 23 \times 3 \\ \hline 69 \end{array}$$

$$\begin{array}{r} 45 \\ 22 \\ \hline 23 \end{array}$$

$$\begin{array}{r} 2 \\ 17 \times 3 \\ \hline 51 \end{array}$$

$$\begin{array}{r} 22 \\ \hline 29 \end{array}$$

But '-' and '÷' are not Satisfies Associative Property in any set.

$$(ie) \quad 2 - (3 - 5) \neq (2 - 3) - 5$$

$$2 - 3 + 5 \neq 2 - 3 - 5$$

$$-1 + 5 \neq 2 - 8$$

$$4 \neq -6$$

$$Now. \quad 2 \div (4 \div 5) \neq (2 \div 4) \div 5$$

$$\frac{2}{4 \div 5} \neq \frac{2 \div 4}{5}$$

$$\frac{2}{4/5} \neq \frac{2/4}{5/1}$$

$$\frac{2 \times 5}{4} \neq \frac{2}{4} \times \frac{1}{5}$$

$$\frac{5}{2} \neq \frac{1}{10}$$

Inference From above 3-Properties:- (Binary/closure, Commutative, Associative) ⑤

We infer that ' $-$ ' and ' $\div$ ' doesn't satisfy any of the above Property, whereas $+$ and $\times$ satisfies all the above,

Therefore, Hereafter, We are going drop our discussion for ' $-$ ' and ' $\div$ ', whereas, We are going to stick with $+$ and $\times$ for Forthcoming Properties.

Identity element

$\forall a \in A, \exists e \in A$, such that $a * e = e * a = a$

then e is called Identity element of A .

i.e. e identifies, whatever the element operate with it.
That is why we call it identity element

Obviously, w.r.t $+$,

0 is the identity element

$$0 + a = a + 0 = a, \forall a \in A.$$

and w.r.t ' $\times$ ' or ' $\cdot$ '

1 is the identity element

$$1 \times a = a \times 1 = a, \forall a \in A$$

Inverse element :-

$\forall a \in A, \exists a^{-1} \in A$ such that $a * a^{-1} = a^{-1} * a = e$
where e is the identity element ^{of A} w.r.t $*$.

then we call

a^{-1} is the inverse of a

(or) a is the inverse of a^{-1} .

Obviously, w.r.t $+$,

$-a$ is the inverse of a .

Since $-a + a = a + (-a) = 0$ (Identity w.r.t $+$)

we call a [or $(-a)$] is a Additive inverse
of $(-a)$ [or (a)]

and w.r.t $\times$,

$\frac{1}{a}$ is the inverse of a

Since $\frac{1}{a} \times a = a \times \frac{1}{a} = 1$ (Identity w.r.t $\times$)

we call, $\frac{1}{a}$ [or a] is a Multiplicative inverse
of a [or $\frac{1}{a}$].

———— $\times$ ———— $\times$ ————

Groups :-

A Set G along with ' $*$ ', is said to be group
($G, *$) if following conditions/axioms are satisfied.

- (i) Closure w.r.t $*$ (ii) Existence of Identity w.r.t $*$
- (iii) Associative w.r.t $*$ (iv) Existence of Inverse element w.r.t $*$.

Abelian group

⑥

A Group $(G, *)$ is said to be Abelian group if $*$ is commutative.

(i.e., $a * b = b * a \quad \forall a, b \in G$.)

— x — x —

For ex:-

1) $(\mathbb{Z}, +)$ is a Abelian group.

2) $(\mathbb{Z}, \cdot)$ is not a (Abelian) group / group.

Since, There is no multiplicative inverse $\frac{1}{a}$ in $\mathbb{Z}$ (Integer set)

3) $(\mathbb{Q}, +)$ is a Abelian group

4) $(\mathbb{Q}, \cdot)$ is not group. / Abelian group.

Since by a single element Zero,

(i.e. Zero doesn't have multiplicative inverse $\frac{1}{0} = \infty$
(∞ is not a definite Number))

∴ If we remove Zero from $\mathbb{Q}$, then we have multiplicative inverse for other elements

5) $(\mathbb{Q} - \{0\}, \cdot)$ is a Abelian group

|||y

6) $(\mathbb{R}, +)$ is a Abelian group

7) $(\mathbb{R} - \{0\}, \cdot)$ is a Abelian group

8) $(\mathbb{C}, +)$ is a Abelian group

9) $(\mathbb{C} - \{0\}, \cdot)$ is a Abelian group.

Field:-

A set F along with '+' and $\cdot$ is said to be Field $(F, +, \cdot)$ if the following conditions are satisfied, They are

- (i) $(F, +)$ is a Abelian group
- (ii) $(F - \{0\}, \cdot)$ is a Abelian group.
- (iii) Distributive property of $\cdot$ over $+$
$$a \cdot (b + c) = a \cdot b + a \cdot c \quad \forall a, b, c \in F$$

Note:-

Third Property (Distributive) is very obvious. Naturally it will satisfies in any usual $\times$ and usual $+$.

So we need to check always the first two conditions

————— $\times$ ————— $\times$ —————

For Ex:-

Having Experience from the Examples of groups

Here we may introduce some Examples for Field.

- (i) $(\mathbb{Z}, +, \cdot)$ is not a field

Since, After removing zero also, $\mathbb{Z}$ is not a group w.r.t $\cdot$.

- (ii) $(\mathbb{Q}, +, \cdot)$ is a Field

- (iii) $(\mathbb{R}, +, \cdot)$ and $(\mathbb{C}, +, \cdot)$ are also Fields.

————— $\times$ —————

Remark 1:-

⑦

In our Inclusion Order of the
Number System,

$$\mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$$

Here except $\mathbb{Z}$ other are Field,

Therefore, ~~Here after~~ Right from Now, We are
going to drop $\mathbb{Z}$ from our discussion, like
How we drop '-' and $\div$.

———— x ——— x ———

Note:-

In the defn of Field, '+' and '·' are not
necessarily our usually Addition and
usual multiplication.

Of course, For the Example of Field.
 $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$, they are our usual addition
and usual multiplication.

Generally '+' and '·' are Just a symbols of ~~some~~
~~Newly defined operations~~ & It's going to be
be defined Newly in some Problem.

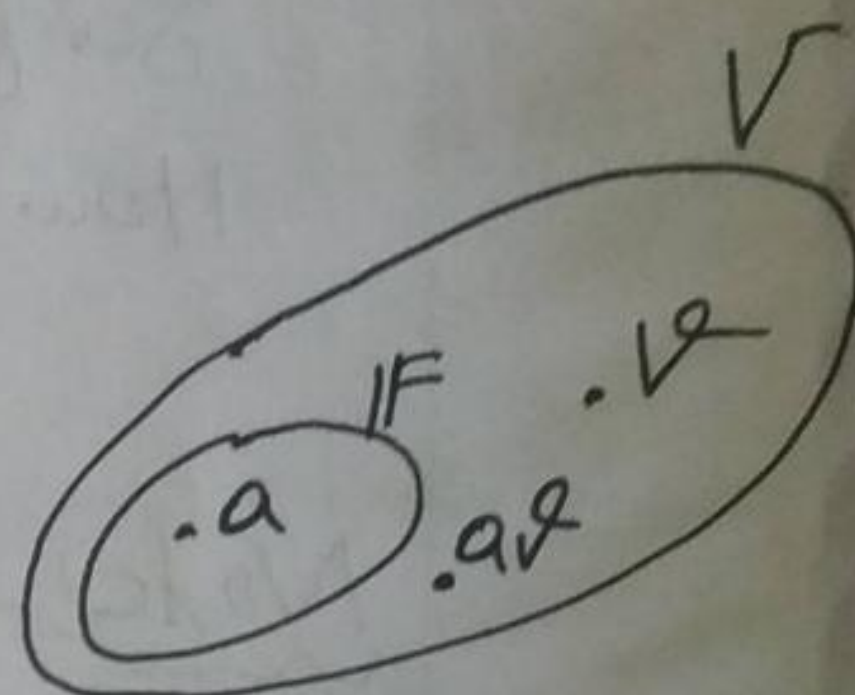
———— x ——— x ———

Vector Space

A set (space) V along with $+$ and scalar multiplication over a Field IF , is said to be Vector Space if the following conditions must be satisfied, They are (i) $(V, +)$ is a Abelian group $[Vs1, \dots, Vs5]$

(Vs6) (ii) closure property of scalar multiplication

$\forall v \in V$, $\forall a \in IF$
 $\downarrow$ $\downarrow$
 we call elements "scalars"
 from V are "vectors"



$\Rightarrow av \in V$
 $\downarrow$
 Idea of usual scalar multiplication $(\underbrace{v + v + \dots + v}_{a \text{ times}})$ clearly $av \in V$ by closure property of $+$
 Thus why, we defined separately.
 [This is because, There may be a Newly defined scalar multiplication.]

(Vs7) $\forall v \in V$, then $1 \cdot v = v$

(Vs8) $\forall v \in V$, $\forall a, b \in IF$, then

$$a(bv) = (ab)v \quad [\text{Bracket chain Prop in scalar mult}]$$

(Vs9) $\forall v \in V$, $\forall a, b \in IF$, then

$$(a+b)v = av + bv \quad [\text{Distributive Prop in vector}]$$

(Vs10) $\forall u, v \in V$, $\forall a \in IF$, then

$$a(u+v) = au + av \quad [\text{Distributive scalar}]$$

~~Theorem 1~~

Cancellation Laws:- (Theorem 1)

LAW 1:- Cancellation law by Additive:- (Right. C. L)

If x, y, z are vectors in a vector space V such that
 $x + z = y + z$, then $x = y$.

Proof:

Given that

V is a vector space

and $x, y, z \in V$ such that

$$x + z = y + z. \quad \text{--- (1)}$$

Now we need to prove: $x = y$

Since $z \in V$, $\exists z^{-1} \in V$ and

Post addition of z^{-1} on both side of (1)

$$(x + z) + z^{-1} = (y + z) + z^{-1}$$

$$x + (z + z^{-1}) = y + (z + z^{-1}) \quad (\because \text{Associative law})$$

$$x + 0 = y + 0 \quad (\because z \text{ is inverse of } z^{-1})$$

$$\boxed{x = y} \quad (0 \text{ is the Identity element of } V)$$

Corollary. 1: The additive identity 0 in a vector space V is unique

Proof:-

Suppose $0_{\in V}$ and $0'_{\in V}$ are two Identity element of V

$$(i) \quad 0 + v = v + 0 = v \quad \forall v \in V$$

$$\& \quad 0' + v = v + 0' = v \quad \text{--- (2)}$$

From (1) & (2)

$$0 + v = 0' + v$$

By cancellation law $\boxed{0 = 0'}$, Hence Identity element of V is unique.

Corollary 2: The additive Inverse of an element in a vector space is unique.

Proof:

For $v \in V$,

~~Assume $v' \in V$ & $v'' \in V$ are two Inverse~~

Assume that $v' \in V$ & $v'' \in V$ are two Inverse element of $v \in V$.

$$(1) \quad v + v' = v' + v = 0 \quad \text{--- (1)}$$

$$\& \quad v + v'' = v'' + v = 0 \quad \text{--- (2)}$$

$$\text{From (1) \& (2)} \quad v + v' = v + v''$$

$$\text{By cancellation law, } \boxed{v' = v''}$$

Hence Inverse of any element is unique

Law 2 (Cancellation law by scalar multiplication)

Let V be a vector space over a field F ,

(i) If $a, b \in F$ and v is a non-zero vector of V .

and $av = bv$, then $\boxed{a = b}$

(ii) If $u, v \in V$, and $a \in F$ be a non-zero vector scalar and

$$au = av \Rightarrow \boxed{u = v}$$

Proof: (i) Given that

$a, b \in F$, $v \in V$ be a non-zero vector

$$\text{and } av = bv$$

$$\Rightarrow av - bv = 0 \quad \left(\begin{array}{l} \text{Post addition by Inverse of } bv \\ \text{i.e., } -bv \end{array} \right)$$

$$\Rightarrow av + (-b)v = 0 \quad (\text{v.s.8})$$

$$\Rightarrow [a + (-b)]v = 0 \quad (\text{v.s.9})$$

$$\Rightarrow a + (-b) = 0 \quad (\because v \neq 0)$$

$$\Rightarrow \boxed{a = b} \quad (\because \text{Post add by } b)$$

(ii) Proof:-

Given that

$$u, v \in V, \text{ and } 0 \neq a \in F$$

$$\text{and } au = av$$

$$\Rightarrow au - av = 0 \quad (\because \text{by Post add by Inverse of } av)$$

$$\Rightarrow \cancel{au} + \cancel{(-a)v} = 0$$

$$\Rightarrow au + a(-v) = 0 \quad (\because \text{VS 8})$$

$$\Rightarrow a[u - v] = 0 \quad (\because \text{VS 10})$$

$$\Rightarrow u - v = 0 \quad (\because a \neq 0)$$

$$\Rightarrow \boxed{u = v} \quad (\because \text{by Post add by Inverse of } -v)$$

_____ x _____ x _____

Properties of Scalar multiplication

In any vector space V , the following statements are true.

(a) $0x = 0$ for $x \in V$

(b) $a0 = 0$ for each $a \in F$

(c) $(-a)x = -(ax) = a(-x)$ For each $a \in F$ and $x \in V$.

if $v = (x, y)$
 Suppose $c(x, y) = (cx, y)$
 then $0v = 0(x, y)$
 $= (0x, y)$
 $= (0, y)$
 SM
 (i.e., $0v \neq 0$
 According to this

Proof:-

(a) Need to prove $0x = 0$

By scalar multiplication $0 \in F, x \in V \Rightarrow 0x \in V$

and 0 is the Identity element in V $\because 0x \in V$

$$\begin{aligned} \therefore 0 + 0x &= 0x \\ &= (0+0)x \quad (\because 0 \text{ is the identity element of } F) \\ &= 0x + 0x \quad (\because \text{VS 9}) \end{aligned}$$

$$\boxed{0 = 0x} \quad \text{by Cancellation law.}$$

(b) To prove that $a0 = 0$, $\forall a \in F$

By scalar multiplication, $a \in F$, $0 \in V$, $\Rightarrow a0 \in V$

and 0 is the identity element of V $\& a0 \in V$

$$a0 + 0 = a0$$

$$= a(0+0)$$

$$= a0 + a0 \quad (\because V \text{ is } \mathbb{A})$$

$$0 = a0 \quad (\because \text{by cancellation law})$$

(c) First, we prove, $(-a)x = -(ax)$

$\forall a \in F$, $x \in V \Rightarrow ax \in V$, $\therefore \exists -(ax) \in V$, s.t.

$$ax + [-(ax)] = 0 \quad \text{--- (1)}$$

By (a)

$$0 = 0x$$

$$= (a + (-a))x$$

$$= ax + (-a)x \quad \text{--- (2)} \quad (\because \text{by distributive law})$$

From (1) & (2)

$$ax + [-(ax)] = ax + (-a)x$$

By cancellation law

$$-(ax) = (-a)x$$

And we need to prove

$$(-a)x = \text{---} (a)(-x)$$

$$(-a)x = [(-1)a]x$$

$$= [a(-1)]x$$

$$= a[(-1)x] \quad (\because \text{by } V_8)$$

$$= a(-x)$$

$$\text{---} x \text{ ---} x \text{ ---}$$

$$\begin{aligned} & \begin{array}{l} -1 \cdot x \\ -(1 \cdot x) \\ -x \end{array} \rightarrow 1 \cdot (-x) = -x \\ & \forall x \in V \end{aligned}$$

Problems -

(10)

- ① Let V denote the set of ~~ordered~~ pairs of real numbers
If (a_1, a_2) and (b_1, b_2) are elements of V and

Example :-

From the Inclusion order of Number system

$$\mathbb{Z} \subseteq \mathbb{Q} \subseteq \mathbb{R} \subseteq \mathbb{C}$$

Except $\mathbb{Z}$ others are Field

- 1) Can $\mathbb{Q}$ be Vector space over $\mathbb{Z}$?

No, since $\mathbb{Z}$ is not a field.

- 2) $\mathbb{Q}$ is a vector space over $\mathbb{Q}$.

Yes, since $\mathbb{Q}$ is a field

- 3) Can $\mathbb{Q}$ be vector space over $\mathbb{R}$?

No, since we can build V.S over a field, which is subset of V.S, but here $\mathbb{Q} \subseteq \mathbb{R}$.

For Example,

Choose scalar $\sqrt{2} \in \mathbb{R}$ and vector $\frac{45}{3} \in \mathbb{Q}$

$$\Rightarrow \sqrt{2} \frac{45}{3} = \frac{\sqrt{2} 45}{3} \notin \mathbb{Q}$$

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- 4) $\mathbb{R}$ is a ^{space} vector space over $\mathbb{R}$

- 5) $\mathbb{R}$ ~~can~~ be a vector space over $\mathbb{Q}$

- 6) $\mathbb{C}$ is a vector space over $\mathbb{C}$

- 7) $\mathbb{C}$ is a vector space over $\mathbb{R}$

- 8) $\mathbb{C}$ is also vector space over $\mathbb{Q}$.

Problems

- ① Let V denote the set of ordered pairs of real numbers. If (a_1, a_2) and (b_1, b_2) are elements of V and $c \in \mathbb{R}$, define

$$(a_1, a_2) + (b_1, b_2) = (a_1 + b_1, a_2 b_2)$$

$\downarrow$ usual +, $\downarrow$ (usual $\times$) in $\mathbb{R}$

$$\text{and } c(a_1, a_2) = (ca_1, a_2)$$

Is V a vector space over $\mathbb{R}$ with these operations?

Justify your answer.

Soln:-

$$\text{Given that } V = \{ (a_1, a_2) \mid a_i \in \mathbb{R}, i=1,2 \}$$

Note that Field is $\mathbb{R}$ which is group w.r.t + and $\mathbb{R} - \{0\}$ is also group w.r.t $\cdot$.

To check (V is a Abelian group w.r.t '+')

VS 1) Closure

$$\text{Let } v_1 = (a_1, a_2) \in V$$

$$v_2 = (b_1, b_2) \in V, \text{ To prove } v_1 + v_2 \in V$$

$$\text{and } v_1 + v_2 = (a_1, a_2) + (b_1, b_2)$$

$$= (a_1 + b_1, a_2 b_2) \in V$$

$\in \mathbb{R} \quad \in \mathbb{R}$

($\because$ Sum and Product of real no are real no)

VS 2) Associative

$$\text{Let } v_3 = (c_1, c_2) \in V$$

To prove

$$v_1 + (v_2 + v_3) = (v_1 + v_2) + v_3$$

$$v_1 + (v_2 + v_3) = (a_1, a_2) + [(b_1, b_2) + (c_1, c_2)]$$

$$= (a_1, a_2) + (b_1 + c_1, b_2 + c_2)$$

$$= (a_1 + (b_1 + c_1), a_2(b_2 + c_2))$$

(+ and . are Associative in $\mathbb{R}$)

$$= ((a_1 + b_1) + c_1, (a_2 b_2) c_2)$$

$$= ((a_1 + b_1), a_2 b_2) + (c_1, c_2)$$

$$= [(a_1, a_2) + (b_1, b_2)] + (c_1, c_2)$$

$$= (v_1 + v_2) + v_3$$

Vs3) Existence of Identity element

Rough work

Let $v_1 = (a_1, a_2) \in V$ for $a_1, a_2 \in \mathbb{R}$

$$(a_1, a_2) + (e_1, e_2) = (a_1, a_2)$$

$\exists e = (0, 1) \in V$ with $0, 1 \in \mathbb{R}$.

$$(a_1 + e_1, a_2 e_2) = (a_1, a_2)$$

Such that

$$\Rightarrow a_1 + e_1 = a_1 \Rightarrow e_1 = 0$$

$$a_2 e_2 = a_2 \Rightarrow e_2 = 1$$

$$\therefore (e_1, e_2) = (0, 1)$$

$$v_1 + e = (a_1, a_2) + (0, 1)$$

$$= (a_1 + 0, a_2 \cdot 1)$$

$$= (a_1, a_2)$$

$$= v_1$$

Hence $e = (0, 1)$ is the Additive identity

— x —

Vs 4) Existence of Inverse:-

Let $v_1 = (a_1, a_2) \in V$, with $a_1, a_2 \in \mathbb{R}$

$\exists v' = (-a_1, \frac{1}{a_2})$ with $-a_1, \frac{1}{a_2} \in \mathbb{R}$

Proof was

$$(a_1, a_2) + (a', a'') = (0, 1) = e$$

$$\Rightarrow (a_1 + a', a_2 a'') = (0, 1)$$

$$\Rightarrow a_1 + a' = 0 \Rightarrow a' = -a_1$$

$$\Rightarrow a_2 a'' = 1 \Rightarrow a'' = \frac{1}{a_2}$$

S.t

$$v_1 + v' = (a_1, a_2) + (-a_1, \frac{1}{a_2})$$

$$= (a_1 + (-a_1), a_2 \frac{1}{a_2})$$

$$= (0, 1)$$

$= e$, Identity element of V

Hence $v' = (-a_1, \frac{1}{a_2})$ is the inverse of $(a_1, a_2) = v_1$

————— x —————

Vs 5) Commutative law

To prove, $v_1 + v_2 = v_2 + v_1$.

$$\text{For, } v_1 + v_2 = (a_1, a_2) + (b_1, b_2)$$

$$= (a_1 + b_1, a_2 b_2)$$

$$= (b_1 + a_1, b_2 a_2)$$

$$= (b_1, b_2) + (a_1, a_2)$$

$$= v_2 + v_1$$

[Since $a_1, a_2 \in \mathbb{R}$, is a field
& $+$ & $\cdot$ satisfy Commutative
law

————— x —————

Vs 6) closure property of scalar multiplication:-

For $v_1 = (a_1, a_2) \in V$, with $a_1, a_2 \in \mathbb{R}$

$\lambda \in \mathbb{R}$ is a field

To prove $c\mathcal{V}_1 \in V$

For, $c\mathcal{V}_1 = c(a_1, a_2)$

$$= (ca_1, ca_2) \in V, \text{ Since } a_1, a_2 \in \mathbb{R}, \& c \in \mathbb{R} \\ \Rightarrow ca_1, ca_2 \in \mathbb{R}$$

Vs7) To prove $1 \cdot \mathcal{V}_1 = \mathcal{V}_1$

$$1 \cdot \mathcal{V}_1 = 1 \cdot (a_1, b_1)$$

$$= (1 \cdot a_1, 1 \cdot b_1)$$

$$= (a_1, b_1) \text{ Since } a_1 \in \mathbb{R}, \& 1 \cdot a_1 = a_1$$

$$= \mathcal{V}_1$$

Vs8) To prove $(ab)\mathcal{V}_1 = a(b\mathcal{V}_1)$ with $a, b \in \mathbb{F}$

Proof:

$$(ab)\mathcal{V}_1 = (ab)(a_1, a_2)$$

$$= ((ab)a_1, a_2)$$

$$= (a(ba_1), a_2) \text{ (Associative in } \mathbb{R})$$

$$= a(ba_1, a_2)$$

$$= a[b(a_1, a_2)]$$

$$= a[b\mathcal{V}_1]$$

Vs9) To prove $(a+b)\mathcal{V}_1 = a\mathcal{V}_1 + b\mathcal{V}_1$ with $a, b \in \mathbb{F}$

Proof:

$$(a+b)\mathcal{V}_1 = (a+b)(a_1, a_2)$$

$$= ((a+b)a_1, a_2)$$

$$= (aa_1 + ba_1, a_2) \text{ (}\because \text{Distributive Property in } \mathbb{R})$$

$$= (aa_1 + ba_1, 1 \cdot a_2)$$

$$= (aa_1, 1) + (ba_1, a_2)$$

$$= a(a_1, 1) + b(a_1, a_2)$$

$$= a(a_1, 1) + b\mathcal{V}_1 \neq a\mathcal{V}_1 + b\mathcal{V}_1$$

Thus V_8 fails,

∴ V is not a vector space over $\mathbb{R}$.

with the newly defined + and scalar multiplication

— x — x —

n-tuple

If $a_1, a_2, \dots, a_n$ are elements of a field $\mathbb{F}$

then $(a_1, a_2, \dots, a_n)$ is called n-tuple

Operations on n-tuples:-

Equality:-

If two n-tuples are equal

$$\text{i.e. } (a_1, a_2, \dots, a_n) = (b_1, b_2, \dots, b_n)$$

$$\text{then } a_i = b_i \quad \forall i = 1, 2, \dots, n$$

Co-ordinate wise Addition of n-tuples

Suppose $u = (a_1, a_2, \dots, a_n)$, $v = (b_1, b_2, \dots, b_n)$ then

$$u + v = (a_1, a_2, \dots, a_n) \overset{\substack{\uparrow \\ \text{co-ordinate wise addition}}}{+} (b_1, b_2, \dots, b_n)$$

$$= (a_1 + b_1, a_2 + b_2, \dots, a_n + b_n)$$

↓
Normal
Addition in $\mathbb{F}$

Co-ordinate wise scalar multiplication of n-tuple:-

If $c \in \mathbb{F}$, $u = (a_1, a_2, \dots, a_n)$

$$cu = c(a_1, a_2, \dots, a_n)$$

$$= (ca_1, ca_2, \dots, ca_n)$$

Note:

The set of all n -tuples with entries from a field F is denoted by F^n .

(i) If $a_1, a_2, \dots, a_n \in F$, then

$$(a_1, a_2, \dots, a_n) \in F^n$$

(ii) If $a_1, a_2 \in \mathbb{R}$, then $(a_1, a_2) \in \mathbb{R}^2$

where $\mathbb{R}$ is the Field of Real numbers. $(3, -2) \in \mathbb{R}^2$

(iii) If $a_1, a_2 \in \mathbb{C}$, then $(a_1, a_2) \in \mathbb{C}^2$,

where $\mathbb{C}$ is the Field of Complex Numbers

$$(1+i, 4i) \in \mathbb{C}^2$$

Problem 1:

(2) Let $V = \mathbb{R}^n = \{(a_1, a_2, \dots, a_n) / a_i \in \mathbb{R}, i=1, 2, \dots, n\}$

Is V a vector space over $\mathbb{R}$ with co-ordinate wise addition and scalar multiplication?

Also check if $V = \mathbb{R}^n$ is a vector space over the field of Complex Numbers with co-ordinate wise addition & ^{scalar} multiplication.

Proof: (sol)

Given that

$$V = \mathbb{R}^n = \{(a_1, a_2, \dots, a_n) / a_i \in \mathbb{R}, i=1, 2, \dots, n\}$$

To check, V is the Abelian group with co-ordinate wise '+'

VS1) Closure wrt co-ordinate wise '+'

Let $u = (a_1, a_2, \dots, a_n) \in V$

$v = (b_1, b_2, \dots, b_n) \in V$

$w = (c_1, c_2, \dots, c_n) \in V$ with $a_i, b_i, c_i \in \mathbb{R}, i=1, 2, \dots, n$

To prove, $u+v \in V$

For, $u+v = (a_1, a_2, \dots, a_n) + (b_1, b_2, \dots, b_n)$

$= (a_1+b_1, a_2+b_2, \dots, a_n+b_n)$

$\in \mathbb{R}^n = V$, Since $a_i+b_i \in \mathbb{R}, i=1, 2, \dots, n$.

VS2) Associative

To prove, $u+(v+w) = (u+v)+w$

Proof:

$u+(v+w) = (a_1, a_2, \dots, a_n) + [(b_1, b_2, \dots, b_n) + (c_1, c_2, \dots, c_n)]$

$= (a_1, a_2, \dots, a_n) + (b_1+c_1, b_2+c_2, \dots, b_n+c_n)$

$= (a_1+(b_1+c_1), a_2+(b_2+c_2), \dots, a_n+(b_n+c_n))$

$= ((a_1+b_1)+c_1, (a_2+b_2)+c_2, \dots, (a_n+b_n)+c_n)$

Associative law is true

in $\mathbb{R}$, since $a_i, b_i, c_i \in \mathbb{R}$

$i=1, 2, \dots, n$

$= (a_1+b_1, a_2+b_2, \dots, a_n+b_n) + (c_1, c_2, \dots, c_n)$

$= [(a_1, a_2, \dots, a_n) + (b_1, b_2, \dots, b_n)] + (c_1, c_2, \dots, c_n)$

$= (u+v)+w$

VS3) Existence of Identity element:-

(14)

For $u \in V$, $\exists O = (0, 0, 0, \dots, 0)_{n \text{ tuple}}$ s.t.
 $O \in \mathbb{R}^n$

$$u + O = (a_1, a_2, \dots, a_n) + (0, 0, \dots, 0)$$

$$= (a_1 + 0, a_2 + 0, \dots, a_n + 0)$$

$$= (a_1, a_2, \dots, a_n) \quad \text{Since } O \in \mathbb{R} \text{ is Identity element.}$$

$$= u$$

VS4) Inverse of Elements:-

For $u \in V$, $\exists \bar{u} = (-a_1, -a_2, -a_3, \dots, -a_n)$ s.t.

$$u + \bar{u} = (a_1, a_2, \dots, a_n) + (-a_1, -a_2, \dots, -a_n)$$

$$= (a_1 - a_1, a_2 - a_2, \dots, a_n + (-a_n))$$

$$= (0, 0, 0, \dots, 0)$$

$$= O \in V = \mathbb{R}^n$$

VS5) Commutative

To Prove $u + v = v + u$

Proof:

$$u + v = (a_1, a_2, \dots, a_n) + (b_1, b_2, \dots, b_n)$$

$$= (a_1 + b_1, a_2 + b_2, \dots, a_n + b_n)$$

$$= (b_1 + a_1, b_2 + a_2, \dots, b_n + a_n), \quad \text{Commutative law is true in } \mathbb{R}$$

$$= (b_1, b_2, \dots, b_n) + (a_1, a_2, \dots, a_n) \quad \text{with } a_i, b_i \in \mathbb{R}$$

$$= v + u.$$

$i = 1, 2, \dots, n$

Vs6) Closure Property of Scalar multiplication:-

For $\forall u \in V, \forall d \in \mathbb{R}$.

To prove, $du \in V = \mathbb{R}^n$.

Proof:-

$$du = d(a_1, a_2, \dots, a_n)$$

$$= (da_1, da_2, \dots, da_n), \text{ Since } d \in \mathbb{R} \text{ \& } a_i \in \mathbb{R}.$$

$$\in \mathbb{R}^n = V$$

Vs7) For $u \in V \Rightarrow 1 \cdot u = u$

Proof

$$1 \cdot u = 1 \cdot (a_1, a_2, \dots, a_n)$$

$$= (1 \cdot a_1, 1 \cdot a_2, \dots, 1 \cdot a_n)$$

$$= (a_1, a_2, \dots, a_n) \quad \left(\begin{array}{l} 1 \text{ is the identity} \\ \text{element of } \mathbb{R} \end{array} \right)$$

$$= u$$

Vs8) $\forall d, e \in \mathbb{R}, u \in V = \mathbb{R}^n$.

To prove $(de)u = d(eu)$

Proof:

$$(de)u = (de)(a_1, a_2, \dots, a_n)$$

$$= ((de)a_1, (de)a_2, \dots, (de)a_n)$$

$$= (d(ea_1), d(ea_2), \dots, d(ea_n))$$

Associative law with $\cdot$ in $\mathbb{R}$ is true

$$= d(ea_1, ea_2, \dots, ea_n)$$

$$(de)a_i = d(ea_i)$$

$$= d[e(a_1, a_2, \dots, a_n)]$$

$$= d[eu]$$

$$(Vs9) \forall d, e \in \mathbb{R}, u \in V = \mathbb{R}^n,$$

$$\text{To prove } (d+e)u = du + eu$$

Proof:

$$\begin{aligned} (d+e)u &= (d+e)(a_1, a_2, \dots, a_n) \\ &= ((d+e)a_1, (d+e)a_2, \dots, (d+e)a_n) \\ &= (da_1 + ea_1, da_2 + ea_2, \dots, da_n + ea_n) \end{aligned}$$

Distributive law is true in $\mathbb{R}$ (Field)

$$\begin{aligned} &= (da_1, da_2, \dots, da_n) + (ea_1, ea_2, \dots, ea_n) \\ &= d(a_1, a_2, \dots, a_n) + e(a_1, a_2, \dots, a_n) \\ &= du + eu. \end{aligned}$$

$$(Vs10) \forall d \in \mathbb{R}, u, v \in \mathbb{R}^n = V$$

$$\text{To prove, } d(u+v) = du + dv$$

Proof:

$$d(u+v) = d[(a_1, a_2, \dots, a_n) + (b_1, b_2, \dots, b_n)]$$

$$= d[a_1 + b_1, a_2 + b_2, \dots, a_n + b_n]$$

$$= (d(a_1 + b_1), d(a_2 + b_2), \dots, d(a_n + b_n))$$

$$= (da_1 + db_1, da_2 + db_2, \dots, da_n + db_n)$$

distributive law is true in $\mathbb{R}$

$$= (da_1, da_2, \dots, da_n) + (db_1, db_2, \dots, db_n)$$

$$\begin{aligned} &= d(a_1, a_2, \dots, a_n) + d(b_1, b_2, \dots, b_n) \\ &= du + dv. \end{aligned}$$

Part II

We need to check $V = \mathbb{R}^n$ is a vector space over $\mathbb{C}$

Straight away, we are going to check,
scalar multiplication;

$$\forall \lambda \in \mathbb{C}, \forall u \in \mathbb{R}^n,$$

To prove, $\lambda u \in \mathbb{R}^n$.

Proof:-

$$\lambda u = \lambda(a_1, a_2, \dots, a_n)$$

$$= (\lambda a_1, \lambda a_2, \dots, \lambda a_n), \quad \begin{array}{l} \lambda \text{ is a complex} \\ a_i \text{'s are Real} \end{array}$$

$$\notin \mathbb{R}^n$$

λa_i 's are complex

$\therefore \mathbb{R}^n$ is not vector space over $\mathbb{C}$

————— x ————— x —————

(H.W) Problem:-

$$\text{Let } V = \mathbb{C}^n = \{(a_1, a_2, \dots, a_n) / a_i \in \mathbb{C}, i=1, 2, \dots, n\}$$

Is V a vector space over $\mathbb{C}$ with usual operation?

Also check if V is a vector space over the field of real numbers with the operation of co-ordinate wise addition & multiplication?

Matrix Notation and operation on matrices:-

Matrices are denoted by capital letters. (Eg A, B, C)

The set of all $m \times n$ matrices with entries from Field F is denoted by $M_{(m \times n)}(F)$

Operations on matrices:-

Matrix addition:

If $A, B \in M_{(m \times n)}(F)$, Then

$$A + B = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} & b_{13} & \dots & b_{1n} \\ b_{21} & b_{22} & b_{23} & \dots & b_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ b_{m1} & b_{m2} & \dots & \dots & b_{mn} \end{bmatrix}$$

Matrix Addition

where a_{ij} 's & b_{ij} 's from F .

$$= \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} & \dots & a_{1n} + b_{1n} \\ a_{21} + b_{21} & a_{22} + b_{22} & \dots & a_{2n} + b_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} + b_{m1} & a_{m2} + b_{m2} & \dots & a_{mn} + b_{mn} \end{bmatrix}$$

Matrix

Scalar multiplication:

ordinary addition in F

If $C \in F$, $A \in M_{(m \times n)}(F)$, Then

$$CA = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} = \begin{bmatrix} Ca_{11} & Ca_{12} & \dots & Ca_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ Ca_{m1} & \dots & \dots & Ca_{mn} \end{bmatrix}$$

Scalar multiplication in F

Problem. Let V denote the set of all $m \times n$ matrices with real entries. Is V a vector space over $\mathbb{R}$ with usual defn of matrix addition & scalar multiplication.

If F is field of rational number, is V a vector space over F ?

Part I:-

$$\text{Given that } M_{(m \times n)}(\mathbb{R}) = V = \left\{ \text{set of } m \times n \text{ matrices with real entries} \right\}$$
$$= \left\{ A \in M_{(m \times n)}(\mathbb{R}) \mid a_{ij} \in \mathbb{R} \right\}$$

(Proof can be done in class room / Home work)

Part II

Is $V =$ set of all $m \times n$ matrices with real entries,
a vector over $\mathbb{F}$ is Field of rational number.

Yes, V is a Vector space over $\mathbb{F}$, since

Product of real & Rational number is always a real number.

Hint Problem:- Let V be the set of sequences $\{a_n\}$ of real numbers. For $\{a_n\}_{n=1}^{\infty}, \{b_n\}_{n=1}^{\infty}$ and any real number t define

$$\{a_n\} + \{b_n\} = \{a_n + b_n\} \quad \left| \quad \{a_n\}_{n=1}^{\infty} = \{a_1, a_2, \dots\} \right.$$
$$\& \quad t \{a_n\} = \{t a_n\}$$

Prove that with these operation V is a Vector space over $\mathbb{R}$.

Problem:-

Let $V = \{ (a_1, a_2) \mid a_1, a_2 \in \mathbb{R} \}$. Define addition of elements of V co-ordinate wise and for

$(a_1, a_2) \in V, \& c \in \mathbb{R}$, define

$$c(a_1, a_2) = \begin{cases} (0, 0) & \text{if } c = 0 \\ (c a_1, c a_2) & \text{if } c \neq 0 \end{cases}$$

Is V a vector space over $\mathbb{R}$ with these operation.

Since, Co-ordinate Addition, always satisfies, Group(A, +) (17)
Property,

Therefore, it is enough to prove, that, V satisfies
Scalar multiplication.

(VS6) $\forall v = (a_1, a_2)$ with $a_1, a_2 \in \mathbb{R} \ \& \ c \in \mathbb{R}$,

Need to prove $cv \in V$

Proof If $c = 0$

$$cv = c(a_1, a_2) \\ = (0, 0) \in \mathbb{R}.$$

If $c \neq 0$.

$$cv = c(a_1, a_2) \\ = (ca_1, \frac{a_2}{c}), \text{ Since } c \in \mathbb{R}, a_1, a_2 \in \mathbb{R} \\ \therefore \frac{a_2}{c} \in \mathbb{R}. \\ \in V$$

(VS7) $\forall v = (a_1, a_2) \in V$,

Need prove $1 \cdot v = v$

Proof

$$1 \cdot v = 1 \cdot (a_1, a_2) \\ = (1 \cdot a_1, \frac{a_2}{1}) \\ = (a_1, a_2) \\ = v$$

(VS8) $\forall a, b \in \mathbb{F}, v = (a_1, a_2) \in V$.

Need to prove $(ab)v = a(bv)$

Proof

$$(ab)v = (ab)(a_1, a_2) \\ = (aba_1, \frac{a_2}{ab})$$

$$= (ab a_1, \frac{a_2}{ab})$$

$$= a (b a_1, \frac{a_2}{b})$$

$$= a [b (a_1, a_2)]$$

$$= a [b v]$$

VS9) $\forall a, b \in \mathbb{F}, v = (a_1, a_2) \in V$

need to prove,

$$(a+b)v = av + bv$$

Proof:

$$(a+b)v = (a+b)(a_1, a_2)$$

$$= ((a+b)a_1, \frac{a_2}{(a+b)})$$

Now

$$av + bv = a(a_1, a_2) + b(a_1, a_2)$$

$$= (aa_1, \frac{a_2}{a}) + (ba_1, \frac{a_2}{b})$$

$$= (aa_1 + ba_1, \frac{a_2}{a} + \frac{a_2}{b})$$

$$= (\cancel{a} + b)a_1, a_2 \left[\frac{1}{a} + \frac{1}{b} \right])$$

$$= ((a+b)a_1, a_2 \left[\frac{a+b}{ab} \right])$$

The 2nd tuple is not equal,

$$\therefore (a+b)v \neq av + bv$$

Thus (VS8) fails.

$\therefore V$ is not a vector space over $\mathbb{R}$

Hue
C.W

Problem:-

(18)

A real valued function f defined on the real line is called an even function if $f(-x) = f(x)$.
for every real number x . Prove that the set of even functions defined on the real line with the operations of addition and scalar multiplication defined as.

$$(f+g)(x) = f(x) + g(x)$$

$$(cf)(x) = cf(x). \text{ For each } x \in \mathbb{R}.$$

$\mathcal{E}$ is a vector space. over $\mathbb{R}$

————— x ————— x —————

Hue Problem:-

show that the set of all real valued polynomials of degree less than (or) equal to n and denoted by $P_n(\mathbb{R})$ where $\mathbb{R}$ is the field of real numbers is a vector space with operations of addition & scalar mult.

Sol:-

Given that

$$V = P_n(\mathbb{R}) = \left\{ u \mid u = a_0 + a_1x + a_2x^2 + \dots + a_nx^n \right. \\ \left. a_i \in \mathbb{R}, i=1,2,\dots,n \right\}$$

First we need prove.

V is a Abelian group w.r.t. $+$

Proof can be done in class/HW/Tutorial

————— x ————— x —————

Soln (for Even function Problem)

Given that $V = \{f(x) \mid f(-x) = f(x), x \in \mathbb{R}\}$.

~~(VS1) Closure~~

Let $f, g, h \in V$, s.t. $f(x) = f(-x)$
 $g(-x) = g(x), h(-x) = h(x)$

(VS1) Closure:-

$\forall f, g \in V$,

To prove $f+g \in V$

Proof:

$$(f+g)(x) = f(x) + g(x)$$

$$\begin{aligned}(f+g)(-x) &= f(-x) + g(-x) \\ &= f(x) + g(x) \\ &= (f+g)(x)\end{aligned}$$

$$\Rightarrow f+g \in V.$$

(VS2) Associative:-

$\forall f, g, h \in V$,

We need to prove $f+(g+h) = (f+g)+h$

Proof:

$$\begin{aligned}[f+(g+h)](x) &= f(x) + (g+h)(x) \\ &= f(x) + g(x) + h(x) \\ &= [f(x) + g(x)] + h(x) \\ &= (f+g)(x) + h(x) \\ &= [(f+g)+h](x)\end{aligned}$$

VS3) Identity:-

(19)

$$\forall f \in V, \nexists z \in V, s.t.$$

$\exists$ a constant function $z(t) = 0$ [Zero function]

$$z(-t) = 0 = z(t).$$

$$\Rightarrow z \in V, s.t.$$

$$(f + z)(t) = f(t) + z(t)$$

$$= f(t) + 0$$

$$= f(t)$$

$$\Rightarrow f + z = f$$

z is a Identity of V .

VS4) Inverse:-

$$\forall f \in V, \nexists g \in V,$$

$\exists -f$, First we need to prove $-f$ is even.

$$(-f)(-t) = -[f(-t)]$$

$$= -f(t), \text{ Since } f \text{ is even}$$

$$= (-f)(t)$$

$$\Rightarrow -f \text{ is even.}$$

$$\Rightarrow -f \in V, \text{ such that}$$

$$[f + (-f)](t) = (f - f)(t)$$

$$= 0$$

$$= z(t), \text{ Identity of } V$$

$$\Rightarrow -f \text{ is the Inverse of } f$$

— x —

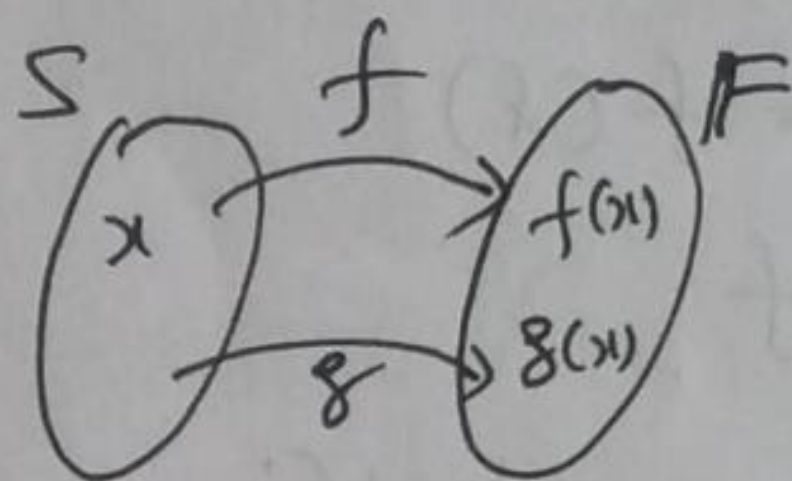
Other Axioms are very easy to prove

— x — x —

Problem

Let S be any non-empty set and let F be any Field. Let V be the set of all functions from S to F .

(1) Let $V = \{f \mid f: S \rightarrow F\}$ (ie, If we define a function from S , then it has a value in F)



Show that V is a vector space over F with the operation defined below.

$$(f+g)(x) = f(x) + g(x), \forall x \in S$$

$$\lambda (cf)(x) = c f(x) \quad \forall x \in S$$

Sol:

VS 1) Closure...

$$\forall f, g \in V \Rightarrow f+g \in V$$

Proof:

$$\begin{aligned} (f+g)(x) &= f(x) + g(x) \\ &= y + z, \text{ where } y \text{ \& } z \text{ are some value of } f \text{ \& } g \text{ in } F \\ &= k \in F \quad k = y + z \in F \Rightarrow \end{aligned}$$

~~We define function from S to F , so k must be~~

$$\Rightarrow f+g \in V.$$

(VS2) Associative

Can be Prove easily.

(20)

(VS3) Identity, Zero function is the Identity, $Z(x) = 0$

(VS4) Inverse!.. $(-f)$ is the Inverse.

Other Axioms are very simple (Can be Prove in class)

————— x —————

Problem!..

Prove that set of all real valued continuous function defined in interval $[0, 1]$ is a vector space over $\mathbb{R}$.

Sol!..

$$V = \left\{ f \mid f: [0, 1] \rightarrow \mathbb{R}, \text{ and } f \text{ is continuous} \right\}$$

Note! Sum of two continuous function is always continuous.

————— x —————

Tutorial / Hm!..

Let V be the set of all pairs (x, y) of real numbers and $\mathbb{R}$ be the field of real numbers. Examine in each of the following cases whether V is a vector space over the field of real numbers or not?

(a) $(x, y) + (x_1, y_1) = (x + x_1, y + y_1)$

$c(x, y) = (|c|x, |c|y)$, (VS¹⁰) Fails, since $|a+b| \neq |a| + |b|$

(b) $(x, y) + (x_1, y_1) = (x + x_1, y + y_1)$

$c(x, y) = (0, cy)$, VS⁷, Fails,

(c) $(x, y) + (x_1, y_1) = (x + x_1, y + y_1)$

$c(x, y) = (c^2x, c^2y)$, (VS¹⁰) Fails, $(a+b)^2 \neq a^2 + b^2$

For (b): - ~~if~~

$$v = (x, y) \in V \Rightarrow 1 \cdot v = v$$

Proof:

$$\begin{aligned} 1 \cdot v &= 1 \cdot (x, y) \\ &= (0, 1 \cdot y) \\ &= (0, y) \\ &\neq (x, y) \\ &\neq v. \end{aligned}$$

———— x ——— x ———

$$\begin{aligned} (d) \quad (x, y) + (x, y) &= (x+x, y+y) \\ C(x, y) &= (x, 0) \end{aligned} \quad \left[\begin{array}{l} \text{VS 1) } 1 \cdot v = 1 \cdot (x, y) \\ = (x, 0) \\ \neq (x, y) \\ \neq v. \end{array} \right]$$

$$\begin{aligned} (e) \quad (x, y) + (x, y) &= (x+2x, x+3y) \\ &\neq C(x, y) = (x, cy). \end{aligned}$$

$$\text{———— x ——— x ——— x ———}$$

$$\mathbb{R}^n = \{ (a_1, a_2, \dots, a_n) / a_i \in \mathbb{R} \}$$

$\mathbb{R}$ is v.s over $\mathbb{R}$
 $\mathbb{R}^2$ is v.s over $\mathbb{R}$
 $\mathbb{R}^3$ is v.s over $\mathbb{R}$

Subspace:-

A subset W of a vector space V over a field F is called a subspace of V if W is a vector space over F with the operation of addition & scalar multiplication defined on V .

$$\mathbb{R} \subseteq \mathbb{R}^2 \subseteq \mathbb{R}^3 \dots \subseteq \mathbb{R}^n$$

Improper subspace (Trivial)

$$2 \leq 2 \quad A \subseteq B \quad (\text{Additive } 0\text{-vector-dense})$$

In any vector space, V and $\{0\}$ are always Subspace of V and are called Improper Subspace.

Zero subspace:- The subspace of V consisting of zero vector only (ie) $\{0\}$ is called zero subspace.

Other subspace of V are called Proper Subspaces.

Theorem 1:-

The necessary & Sufficient Conditions for a non-empty subset W of a vector space V over a field F to be a subspace of V are.

- (i) $0 \in W$
 (ii) $a \in F, u, v \in W \Rightarrow au + v \in W$
- (S.M. closure) (Additive closure)*
- A nonempty subset W of a V.S V over a field F is a subspace of V iff
 (i) $0 \in W$
 (ii) $a \in F, u, v \in W \Rightarrow au + v \in W$

Nec condition:-

Given that

W is a subspace of $V(F)$

Proof:

To prove that (i) $0 \in W$

(ii) $a \in F, u, v \in W \Rightarrow au + v \in W$

Proof (i)

As W is a subspace of $V(F)$,

W satisfies all the axioms of vector space

$\therefore$ Additive identity 0 must be in W

(i.e. $0 \in W$)

Proof for (ii)

$a \in F, u, v \in W \Rightarrow au \in W$ & $v \in W$ (Scalar multi true in W)
 $\Rightarrow au + v \in W$ ($\because W$ is closed under '+')

Hence Nec condition Proved

Sufficient condition:-

Given that (i) $0 \in W$

(ii) $a \in F, u, v \in W \Rightarrow au + v \in W$

To prove that, W is subspace of $V(F)$

(i.e. W is a vector space over F)

It is enough to prove only three axioms

^{closure} (VS1), ^{Inverse} (VS4) and (VS6)SM

Since VS₂ - Associative is true for every vector in V

VS₅ - Commutative is true for every vector in V

VS₇ - $1 \cdot v = v \quad \forall v \in V$

$\left. \begin{matrix} \text{VS}_8 \\ \text{VS}_9 \\ \text{VS}_{10} \end{matrix} \right\}$ - true for every vector in V

$\therefore$ They also hold good for vector of W , since $W \subseteq V$
and (VS₃) is directly hold from Hypothesis (i)

VS₁) $\forall u, v \in W$

To prove $u+v \in W$

Proof:- $1 \in F$, $u, v \in W$, since Hypothesis (ii)

Multiplicative Identity

$$1 \cdot u + v \in W$$

$$\Rightarrow u + v \in W.$$

Hence W is closed w.r.t '+'

~~VS₄) $\forall u \in W$, we need prove of Inverse in W .~~

~~For $-1 \in F$, (Additive inverse of 1,
Since F is a field)~~

~~$u, u \in W$, since by Hypothesis (ii)~~

~~$(-1)u + u \in W$~~

(VS₆) Closure Property of Scalar multiplication.

$\forall a \in F, \forall u \in W,$

To prove that $au \in W$.

Proof:-

$\forall a \in F, u, 0 \in W$, Since by Hypothesis (i) and (ii)

$$au + 0 \in W$$

$$\Rightarrow au \in W$$

Hence scalar multiplication is closed.

(V4) Inverse

For $-1 \in F$ (Inverse of $1 \in F$, is Field)

$\& u \in W$, since by scalar multiplication is closed.

$$(-1)u \in W$$

$$\Rightarrow -u \in W, \text{ such that}$$

$$\Rightarrow u + (-u) = 0$$

Thus $-u$ is the Additive inverse of u exists in W

Hence, all the axioms of vector space hold for W .

$\therefore W$ is Vector space $\& W$ is subset of V

$\Rightarrow W$ is subspace of $V(F)$.

————— $\times$ ————— $\times$ —————

Algebra of Subspace

$$A \cup B = \{x \mid x \in B \text{ or } x \in A\}$$

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$$

$$A' = \{x \in U \mid x \notin A\} = U - A.$$

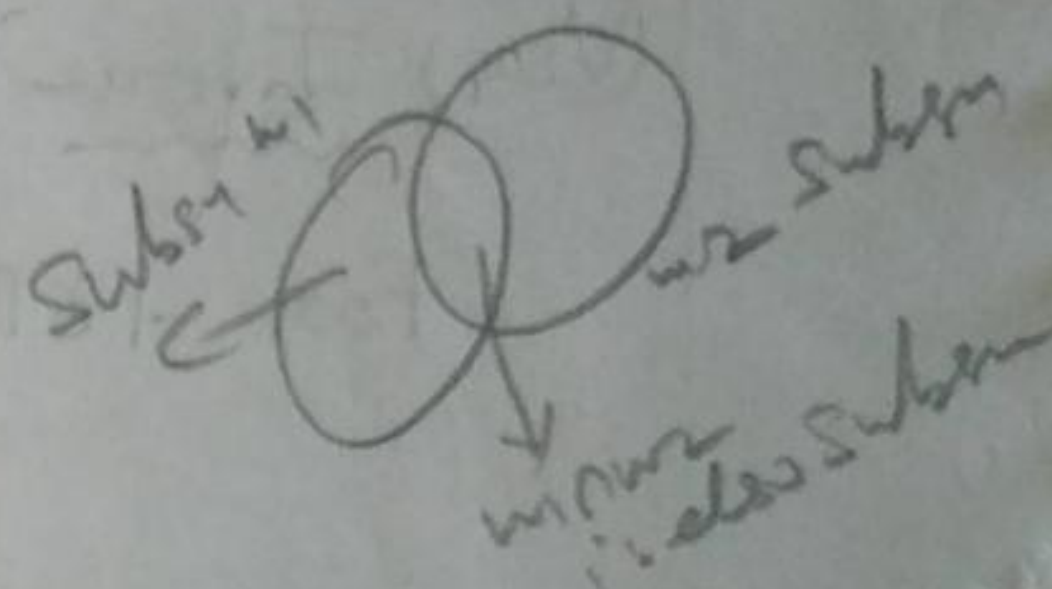
Theorem 2 :-

The intersection of any two subspace W_1 and W_2 of a vector space $V(F)$ is also a subspace of $V(F)$

Proof:

Given that W_1 and W_2 are subspace of vector space $V(F)$.

By nec/suff condition, we have



For W_1 ,

$$(a) 0 \in W_1 \text{ --- } \textcircled{1}$$

$$(b) a \in F, u, v \in W_1$$

$$\Rightarrow au + v \in W_1 \text{ --- } \textcircled{2}$$

For W_2 ,

$$(a) 0 \in W_2 \text{ --- } \textcircled{3}$$

$$(b) a \in F, u, v \in W_2$$

$$\Rightarrow au + v \in W_2 \text{ --- } \textcircled{4}$$

We need to prove,

$W_1 \cap W_2$ is also subspace of $V(F)$

By nec/suff condition, we need to prove

$$(i) 0 \in W_1 \cap W_2$$

$$(ii) a \in F, u, v \in W_1 \cap W_2$$

$$\Rightarrow au + v \in W_1 \cap W_2$$

Proof: for (i)

Since by $\textcircled{1}$ and $\textcircled{3}$, we have

$$0 \in W_1 \text{ and } 0 \in W_2 \Rightarrow 0 \in W_1 \cap W_2$$

Thus (i) proved

For (ii),

$$\forall a \in F, u, v \in W_1 \cap W_2$$

$$\Rightarrow u, v \in W_1 \text{ and } u, v \in W_2$$

$$\Rightarrow au + v \in W_1 \text{ and } au + v \in W_2 \text{ (since by } \textcircled{2} \text{ \& } \textcircled{4})$$

$$\Rightarrow au + v \in W_1 \cap W_2$$

Thus (ii) proved

Theorem 3

(23)

The union of two Subspace is a Subspace
if and only if

one is contained in the other.

(i.e) If W_1 and W_2 are Subspace of V , then $W_1 \cup W_2$ is a Subspace of V if and only if

$$W_1 \subseteq W_2 \text{ (or) } W_2 \subseteq W_1.$$

Proof:-

Part I

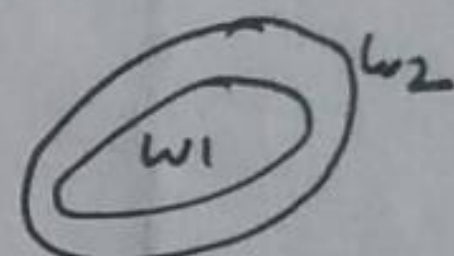
Given that (i) W_1 and W_2 are Subspace of V (IF)

$$(ii) W_1 \subseteq W_2 \text{ (or) } W_2 \subseteq W_1$$

To prove that: $W_1 \cup W_2$ is a Subspace of V (IF).

Proof:

From (ii) $W_1 \subseteq W_2 \Rightarrow W_1 \cup W_2 = W_2$ is a Subspace



or $W_2 \subseteq W_1 \Rightarrow W_1 \cup W_2 = W_1$ is a Subspace (Since by (i))

~~As W_1 & W_2 are Subspace of V (From (i))~~

~~$W_1 \cup W_2$~~

$\Rightarrow W_1 \cup W_2$ is also Subspace of V (IF).

— x —

Part II:-

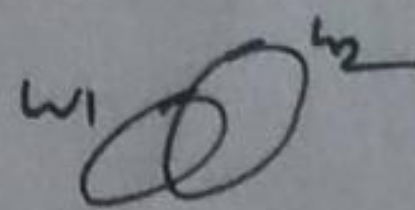
Given that (i) W_1 and W_2 are Subspace of V

(ii) $W_1 \cup W_2$ is a Subspace of V

To prove that: $W_1 \subseteq W_2$ (or) $W_2 \subseteq W_1$

Proof by Contradiction.

Assume that $W_1 \not\subseteq W_2$ and $W_2 \not\subseteq W_1$



$$\therefore W_1 \not\subseteq W_2 \Rightarrow \exists u_1 \in W_1 \text{ and } u_1 \notin W_2 \quad \text{--- ①}$$

$$\& W_2 \not\subseteq W_1 \Rightarrow \exists u_2 \in W_2 \text{ and } u_2 \notin W_1 \quad \text{--- ②}$$

From the above,

$$u_1 \in W_1 \cup W_2 \text{ \& } u_2 \in W_1 \cup W_2$$

and $W_1 \cup W_2$ is a subspace, from hypothesis.

$$\Rightarrow u_1 + u_2 \in W_1 \cup W_2$$

$$\Rightarrow u_1 + u_2 \in W_1 \text{ (or) } u_1 + u_2 \in W_2$$

Case (i): -

Suppose $u_1 + u_2 \in W_1$ and $u_1 \in W_1$ (o.o by ①)

$$\Rightarrow -u_1 \in W_1 \text{ (o.o } W_1 \text{ is subspace)}$$

$$\text{Now } (u_1 + u_2) + (-u_1) \in W_1$$

$$\Rightarrow u_2 \in W_1, \text{ Contradict to ②.}$$

Case (ii): -

Suppose $u_1 + u_2 \in W_2$ and $u_2 \in W_2$ (o.o by ②)

$$\Rightarrow -u_2 \in W_2 \text{ (o.o } W_2 \text{ is subspace)}$$

$$\text{Now } (u_1 + u_2) + (-u_2) \in W_2$$

$$\Rightarrow u_1 \in W_2, \text{ Contradict to ①}$$

$\therefore$ our assumption is wrong,

$$\text{(ie, } W_1 \subseteq W_2 \text{ (or) } W_2 \subseteq W_1$$

— x — x —

Defn Linear Sum of Subsets of a V.S

(24)

If S_1 and S_2 are non-empty subsets of a vector space V , then the sum of S_1 and S_2 denoted by

$$S_1 + S_2 = \{ x + y \mid x \in S_1, y \in S_2 \}$$

Theorem 4:-

Let W_1 and W_2 be two subspaces of V

1) Prove that $W_1 + W_2$ is a subspace of V

that contains both W_1 and W_2

2) Prove that any subspace of V that contains W_1 and W_2 must also contain $W_1 + W_2$

Sol:-

1) Proof

Given that W_1 and W_2 are subspaces of V

By nec/suff condition

$$(a) \quad 0 \in W_1 \text{ and } 0 \in W_2 \quad \text{--- (1)}$$

$$(b) \quad a \in \mathbb{F}, u_1, v_1 \in W_1 \Rightarrow au_1 + v_1 \in W_1 \quad \text{--- (2)}$$

$$a \in \mathbb{F}, u_2, v_2 \in W_2 \Rightarrow au_2 + v_2 \in W_2 \quad \text{--- (3)}$$

To prove that:

(i) $W_1 + W_2$ is a subspace

(ii) $W_1 \subseteq W_1 + W_2$ and $W_2 \subseteq W_1 + W_2$

Proof

(For (i)) We need to show

$$1) \quad 0 \in W_1 + W_2$$

$$2) \quad a \in \mathbb{F}, u, v \in W_1 + W_2$$

$$au + v \in W_1 + W_2$$

Proof For 1)

Since W_1 and W_2 are subspace of V .

we have $0 \in W_1$ & $0 \in W_2$

$$\text{Then } 0+0=0$$

$$\therefore 0 \in W_1 + W_2$$

Proof For 2)

For $a \in F$, $u, v \in W_1 + W_2$

$$\text{Now } u = x_1 + x_2, \quad x_1 \in W_1, \quad x_2 \in W_2 \quad \left(\begin{array}{l} \text{defn of} \\ W_1 + W_2 \end{array} \right)$$
$$\& \quad v = y_1 + y_2, \quad y_1 \in W_1, \quad y_2 \in W_2$$

Consider

$$au + v = a(x_1 + x_2) + (y_1 + y_2)$$

$$= (ax_1 + y_1) + (ax_2 + y_2), \quad \text{with } ax_1 + y_1 \in W_1$$
$$\in W_1 + W_2 \quad \quad \quad ax_2 + y_2 \in W_2$$

Thus $W_1 + W_2$ is a subspace of V

_____ x _____ x _____.

Part 2.

Theorem

The linear sum of 2-subspaces of vector space V over F is a subspace that contain both W_1 and W_2

(24)

Proof:

Given that

V is a vector space over F

and W_1 and W_2 are 2-subspaces of V over F

①

We need to prove: ① $W_1 + W_2$ is a subspace of V

② $W_1 \subset W_1 + W_2$ & $W_2 \subset W_1 + W_2$

Proof: For ①

(i) $0 \in W_1$ — ②

(i) $0 \in W_2$ — ④

Since by ①, (ii) $a \in F, u, v \in W_1$

$\Rightarrow au + v \in W_1$

(ii) $a \in F, u_2, v_2 \in W_2$

$\Rightarrow au_2 + v_2 \in W_2$

⑤

To prove: $W_1 + W_2$ is a subspace, & $W_1 + W_2 = \{x+y \mid x \in W_1, y \in W_2\}$

~~$a \in F, u, v \in W_1 + W_2 \Rightarrow u = x_1 + x_2$~~

~~$v = y_1 + y_2$, where $x_1 \in W_1$~~

~~$x_2, y_1 \in W_1$~~

~~$y_2 \in W_2$~~

~~Since by~~

For that, we need to prove, (i) $0 \in W_1 + W_2$

(ii) $a \in F, u, v \in W_1 + W_2$

$\Rightarrow au + v \in W_1 + W_2$

For (i):

Since by ② & ④

$0 = 0 + 0 \in W_1 + W_2$, where $0 \in W_1, 0 \in W_2$

$\Rightarrow 0 \in W_1 + W_2$

For (ii):

$$a \in F, u, v \in W_1 + W_2 \Rightarrow u = x_1 + x_2 \\ v = y_1 + y_2, \text{ where } x_1, y_1 \in W_1 \\ x_2, y_2 \in W_2$$

~~Since~~ by

$$\begin{array}{l|l} \text{Now } a \in F, x_1, y_1 \in W_1 \text{ (By (3))} & a \in F, x_2, y_2 \in W_2 \text{ (By (4))} \\ \hline ax_1 + y_1 \in W_1 \text{ --- (6)} & ax_2 + y_2 \in W_2 \text{ --- (7)} \end{array}$$

$$\text{Now } au + v = a(x_1 + x_2) + (y_1 + y_2)$$

$$= ax_1 + ax_2 + y_1 + y_2$$

$$= (ax_1 + y_1) + (ax_2 + y_2)$$

$$\in W_1 + W_2 \text{ } \therefore$$

Hence $W_1 + W_2$ is a Subspace of V

Part 2:-

$$W_1 \subset W_1 + W_2 \text{ and } W_2 \subset W_1 + W_2$$

Let $x \in W_1$, we need to prove $x \in W_1 + W_2$

$$\text{Since } x \in W_1 \text{ and } 0 \in W_2 \Rightarrow x + 0 \in W_1 + W_2$$

$$\Rightarrow x \in W_1 + W_2$$

* Hence $W_1 \subset W_1 + W_2$

$$\text{Ily } W_2 \subset W_1 + W_2$$

— x —

Problem:-

- ① Determine whether the following sets are Subspace of $\mathbb{R}^3$ under the operation of addition & Scalar multiplication defined on $\mathbb{R}^3$. Justify your answers.

(a) $W_1 = \left\{ (a_1, a_2, a_3) \in \mathbb{R}^3 / a_1 = 3a_2 \text{ \& } a_3 = -a_2 \right\}$ (is a subspace) (26)

(b) $W_2 = \left\{ (a_1, a_2, a_3) \in \mathbb{R}^3 / a_1 + 2a_2 - 3a_3 = 1 \right\}$ (not a subspace)

(c) $W_3 = \left\{ (a, b, c) \in \mathbb{R}^3 / a + b + 2c = 0 \right\}$ (is a subspace)

Proof/Sol

(a) Given that $W_1 = \left\{ (a_1, a_2, a_3) / a_1 = 3a_2 \text{ and } a_3 = -a_2 \right\}$

Let $u, v \in W_1 \Rightarrow u = (a_1, a_2, a_3)$ s.t. $a_1 = 3a_2, a_3 = -a_2$ (1)

$v = (b_1, b_2, b_3)$ s.t. $b_1 = 3b_2, b_3 = -b_2$ (2)

Check
Need to prove :- W_1 is subspace (or) not.

For (i) $[0 \in W_1]$

As $0 \in \mathbb{R}$, ~~we can~~ $\bar{0} = (0, 0, 0) \in W_1$ s.t.

$0 = 3 \cdot 0, 0 = -0$

$\therefore \boxed{\bar{0} \in W}$

For (ii) :-

$a \in \mathbb{R}, u, v \in W_1 \Rightarrow$

Proof

$au + v = a(a_1, a_2, a_3) + (b_1, b_2, b_3)$

$= (aa_1, aa_2, aa_3) + (b_1, b_2, b_3)$

$= (aa_1 + b_1, aa_2 + b_2, aa_3 + b_3)$

$au + v$ will be a member of W_1 only if

$aa_1 + b_1 = 3[aa_2 + b_2]$

$aa_3 + b_3 = -[aa_2 + b_2]$

Verification of done

$$\begin{aligned} a a_1 + b_1 &= a[3a_2] + 3b_2 \\ &= 3aa_2 + 3b_2 \\ &= 3[a a_2 + b_2] \end{aligned}$$

$$\begin{aligned} \text{By } a a_3 + b_3 &= a[-a_2] + (-b_2) \\ &= -[a a_2 + b_2] \end{aligned}$$

Hence verified

$$\therefore a u + v \in W_1$$

Hence W_1 is a subspace of $\mathbb{R}^3$

Home work (b) & (c)

- ② Let $\mathbb{R}$ be the field of real numbers. Which of the following are subspace of $\mathbb{R}^3$?
- (a) $\{(x, 2y, 3z) \in \mathbb{R}^3 / x, y, z \in \mathbb{R}\} = W_1$ (is a subspace)
 - (b) $\{(x, x, x) / x \in \mathbb{R}\} = W_2$ (is a subspace)
 - (c) $\{(x, y, z) / x, y, z \text{ are rational numbers}\} = W_3$ (not a subspace)

Proof:

$$\begin{aligned} \text{(c) } W_3 &= \{(x, y, z) / x, y, z \text{ are Rational numbers}\} \\ &= \{(x_1, x_2, x_3) / x_i = \frac{p}{q}, q \neq 0, \} \end{aligned}$$

For (i)

(27)

As $0 \in \mathbb{R}$ & $\frac{0}{q} = 0$ any $q \neq 0$

we construct $\bar{0} = (0, 0, 0)$ where $0 = \frac{0}{q}$, for any $q \neq 0$.

Hence $\bar{0} \in W_3$.

For (ii):-

$a \in \mathbb{F} = \mathbb{R}$, a real number, & $u, v \in W_3$ with

$u = (x_1, x_2, x_3)$, where $x_i = \frac{p}{q}$, $q \neq 0$

$v = (y_1, y_2, y_3)$, where $y_i = \frac{p}{q}$, $q \neq 0$.

$$\text{Now } au + v = a(x_1, x_2, x_3) + (y_1, y_2, y_3)$$

$$= (ax_1, ax_2, ax_3) + (y_1, y_2, y_3)$$

$$= (ax_1 + y_1, ax_2 + y_2, ax_3 + y_3)$$

But There is no guarantee that

$ax_i + y_i$ is a Rational Number, $1 \leq i \leq 3$, since $a \in \mathbb{R}$

For:-

$$a = \sqrt{2} \in \mathbb{R}, \quad x_i, y_i \in \mathbb{Q}$$

$$\text{as } \sqrt{2} \quad ax_i + y_i = \sqrt{2} x_i + y_i \neq \frac{p}{q} \quad \text{since } \sqrt{2} \neq \frac{p}{q} \\ \in \mathbb{Q}.$$

$$\therefore au + v \notin W_3, \quad \text{since } ax_i + y_i \neq \frac{p}{q}, \quad q \neq 0.$$

$\Rightarrow W_3$ is not Subspace of V

Other subdivision (a) & (b) are Homework

— x —

The Number 4 can be written as such a way

$$4 = \underset{\alpha}{51}(4076) + \underset{\beta}{(-203)}1024.$$

It's a Problem of Number Theory, that

GCD of (4076, 1024) and 4 can be written as
linear combination of 4076, 1024.

But it is very hard to predict the value of α and β
For what α and β , we write 4 as linear
combination of 4076, 1024.

For ordinary Numbers itself it very difficult to predict
the ~~status~~, Right scalar for linear combination of other
two ~~vectors~~ numbers.

Think how about, vectors.

Can we write,

$(-2, 0, 3)$ as linear combination of $(1, 3, 0)$ & $(2, 4, -1)$
 $\vec{v}$ $\vec{u}_1$ $\vec{u}_2$

If ~~so~~ ^{yes}, what is the scalar a_1 & a_2

$$(-2, 0, 3) = a_1(1, 3, 0) + a_2(2, 4, -1)$$

To find a_1 & a_2 , Proceed Further.

$$\begin{aligned} (-2, 0, 3) &= (a_1, 3a_1, 0) + (2a_2, 4a_2, -a_2) \\ &= (a_1 + 2a_2, 3a_1 + 4a_2, -a_2) \end{aligned}$$

$$\Rightarrow \left. \begin{aligned} a_1 + 2a_2 &= -2 \\ 3a_1 + 4a_2 &= 0 \\ -a_2 &= 3 \end{aligned} \right\} \text{--- (1)}$$

Now the problem of determining whether (or) not a vector can be expressed as a linear combination of other vectors, reduces to the problem of solving a system of linear combination equations.

We know, 'How to solve system of linear equations, by using elimination (Gauss method) method, But here we are going to introduce Gauss elimination method to solve system of linear eqn.

Step 1 :- Write given system in the form of matrix

$$AX = B, \text{ where } X \text{ is the solution column matrix}$$

Step 2 :- Construct the augmented matrix

$$[A|B]$$

Step 3 :- By Row elementary operation, Reduce

$[A|B]$ into Row echelon matrix.

Step 4 :-

If a row is obtained in which only one non-zero entry lies in the last column.

(i) A zero row is obtained in which, a corresponding element in B is non-zero, then conclude that the original system is inconsistent (solution doesn't exist)

(ii) Vector v , can't be expressed as linear combination of other vectors

Else, if A zero row is obtained in A, and corresponding element in B is also zero, then discard the zero row completely and rewrite the new system of equations. (29)

Elementary Row Operation:-

- 1) Interchanging of two rows (is allowed) $[R_i \leftrightarrow R_j]$
- 2) Multiplying any row of A by a non-zero scalar $[R_i \rightarrow kR_i] \quad k \neq 0$
 $\downarrow$ scalar multiple row
- 3) adding any scalar multiple of a row of A to another row $R_i \rightarrow R_i + kR_j$
 Replacing a row by adding some other other row with it.

Echelon Form:-

In any matrix, Entries of each column below a_{ii} is zero, then it is called Echelon matrix.

$$\begin{bmatrix} 2 & 3 & 4 \\ 0 & 4 & 0 \\ 0 & 0 & 2 \end{bmatrix} \quad \begin{bmatrix} 4 & 5 & 6 & 7 \\ 0 & 8 & 7 & 9 \\ 0 & 0 & 6 & 7 \\ 0 & 0 & 0 & 9 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 2 & 3 & 5 \\ 0 & 4 & 7 & 9 & 10 \\ 0 & 0 & 8 & 1 & 2 \end{bmatrix}$$

Let us continue the problem.

System (1) $\Rightarrow$

Step 1:

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} -2 \\ 0 \\ 3 \end{bmatrix}$$

$A \quad \times \quad B$

Step 2

Augmented matrix

$$[A|B] \sim \left[\begin{array}{cc|c} 1 & 2 & -2 \\ 3 & 4 & 0 \\ 0 & -1 & 3 \end{array} \right] R_2 \rightarrow R_2 - 3R_1$$

$$\sim \left[\begin{array}{cc|c} 1 & 2 & -2 \\ 0 & -2 & 6 \\ 0 & -1 & 3 \end{array} \right] R_2 \rightarrow \frac{R_2}{-2}$$

$$\sim \left[\begin{array}{cc|c} 1 & 2 & -2 \\ 0 & 1 & -3 \\ 0 & -1 & 3 \end{array} \right] R_3 \rightarrow R_3 + R_2$$

$$\sim \left[\begin{array}{cc|c} 1 & 2 & -2 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \end{array} \right]$$

Discarding zero row completely

$$\sim \left[\begin{array}{cc|c} 1 & 2 & -2 \\ 0 & 1 & -3 \end{array} \right]$$

Bring back to system

$$a_1 + 2a_2 = -2$$

$$a_2 = -3$$

$$\text{Now } a_1 = -2 - 2(-3)$$

$$= -2 + 6$$

$$= +4$$

$$\boxed{a_1 = 4, a_2 = -3}$$

Now

$$(-2, 0, 3) = 4(1, 3, 0) +$$

Now

$$\begin{aligned} 4(1, 3, 0) + (-3)(2, 4, -1) &= (4, 12, 0) + (-6, -12, +3) \\ &= (4-6, 12-12, 0+3) \\ &= (-2, 0, 3) \end{aligned}$$

Thus,

$$(-2, 0, 3) = 4(1, 3, 0) + (-3)(2, 4, -1)$$

— X —

Problem 2:

Check if $(2, 6, 8)$ can be expressed as linear combination of $(1, 2, 1)$, $(-2, -4, -2)$, $(0, 2, 3)$, $(2, 0, -3)$, $(-3, 8, 16)$

Sol:

$\exists$ scalar $a_1, a_2, a_3, \dots, a_5$ such that

$$\begin{aligned} (2, 6, 8) &= a_1(1, 2, 1) + a_2(-2, -4, -2) + a_3(0, 2, 3) + a_4(2, 0, -3) + a_5(-3, 8, 16) \\ &= (a_1, 2a_1, a_1) + (-2a_2, -4a_2, -2a_2) + (0, 2a_3, 3a_3) + (2a_4, 0, -3a_4) \\ &\quad + (-3a_5, 8a_5, 16a_5) \\ &= (a_1 - 2a_2 + 2a_4 - 3a_5, 2a_1 - 4a_2 + 2a_3 + 8a_5, a_1 - 2a_2 + 3a_3 - 3a_4 + 16a_5) \end{aligned}$$

$$\Rightarrow \left. \begin{aligned} a_1 - 2a_2 + 2a_4 - 3a_5 &= 2 \\ 2a_1 - 4a_2 + 2a_3 + 8a_5 &= 6 \\ a_1 - 2a_2 + 3a_3 - 3a_4 + 16a_5 &= 8 \end{aligned} \right\} \text{--- (1)}$$

System into matrix

$$\Rightarrow \begin{bmatrix} 1 & -2 & 0 & 2 & -3 \\ 2 & -4 & 2 & 0 & 8 \\ 1 & -2 & 3 & -3 & 16 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \end{bmatrix} = \begin{bmatrix} 2 \\ 6 \\ 8 \end{bmatrix}$$

A x B

Step 2.

Augmented matrix is

$$[A|B] \sim \begin{bmatrix} 1 & -2 & 0 & 2 & -3 & 2 \\ 2 & -4 & 2 & 0 & 8 & 6 \\ 1 & -2 & 3 & -3 & 16 & 8 \end{bmatrix} \quad R_2 \rightarrow \frac{R_2}{2}$$

$$\sim \begin{bmatrix} 1 & -2 & 0 & 2 & -3 & 2 \\ 1 & -2 & 1 & 0 & 4 & 3 \\ 1 & -2 & 3 & -3 & 16 & 8 \end{bmatrix} \quad \begin{array}{l} R_2 \rightarrow R_2 - R_1 \\ R_3 \rightarrow R_3 - R_1 \end{array}$$

$$\sim \begin{bmatrix} 1 & -2 & 0 & 2 & -3 & 2 \\ 0 & 0 & 1 & -2 & 7 & 1 \\ 0 & 0 & 3 & -5 & 19 & 6 \end{bmatrix} \quad \begin{array}{l} R. \\ R_3 \rightarrow R_3 - 3R_2 \end{array}$$

$$\sim \begin{bmatrix} 1 & -2 & 0 & 2 & -3 & 2 \\ 0 & 0 & 1 & -2 & 7 & 1 \\ 0 & 0 & 0 & 1 & -2 & 3 \end{bmatrix} \quad \boxed{R_3 \rightarrow 2R_3 - R_2}$$

$$\sim \begin{bmatrix} 1 & -2 & 0 & 2 & -3 & 2 \\ 0 & 0 & 1 & 2 & 7 & 1 \\ 0 & 0 & 0 & -1 & -1 & -1 \end{bmatrix}$$

Bring back to system

$$a_1 - 2a_2 + 2a_4 - 3a_5 = 2$$

$$a_3 + 2a_4 + 7a_5 = 1$$

$$a_4 - 2a_5 = 3$$

Consider a_5 as Free Variable

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$$\therefore \text{Put } \boxed{a_5 = 0}$$

$$\boxed{a_4 = 3}$$

$$a_3 = 1 + 2(3) - 7(0) \\ = 1 + 6$$

$$\boxed{a_3 = 7}$$

$$a_1 - 2a_2 = 2 - 2(3) + 3(0)$$

$$a_1 - 2a_2 = 2 - 6 \\ = -4$$

$$\text{Put } \boxed{a_2 = 0}$$

$$\boxed{a_1 = -4}$$

$$\therefore (a_1, a_2, a_3, a_4, a_5) = (-4, 0, 7, 3, 0)$$

$$-4(1, 2, 1) + 0(-2, -4, -2) + (7)(0, 2, 3) + 3(2, 0, -3) + 0(-3, 8, 6)$$

$$= (-4, -8, -4) + (0, 14, 21) + (6, 0, -9)$$

$$= (-4 + 6, -8 + 14, -4 + 21 - 9)$$

$$= (2, 6, 8)$$

Hence

$(2, 6, 8)$ is a linear combination of given five vectors

— x — x —

Homework 1:- Find each of the following list of vectors in $\mathbb{R}^3$, determine whether the first vector can be expressed as a linear combination of the other two

(b) $(3, 4, 1), \{(1, -2, 1), (-2, -1, 1)\}$ Inconsistent

(c) $(-2, 2, 2) \{(1, 2, -1), (-3, -3, 3)\}$ $a_1 = 4, a_2 = 2$

×

Polynomial Problems:- For each list of Polynomials in $\mathbb{P}_3(\mathbb{R})$ determine whether the first Polynomial can be expressed as a linear combination of the other two

H.W (a) $-2x^3 - 11x^2 + 3x + 2, x^3 - 2x^2 + 3x - 1, 2x^3 + x^2 + 3x - 2$

C.W (b) $3x^3 - 2x^2 + 7x + 8, x^3 - 2x^2 - 5x - 3, 3x^3 - 5x^2 - 4x - 9$

Sol:-

$$\begin{aligned} (b) \quad 3x^3 - 2x^2 + 7x + 8 &= a_1(x^3 - 2x^2 - 5x - 3) + a_2(3x^3 - 5x^2 - 4x - 9) \\ &= a_1x^3 - 2a_1x^2 - 5a_1x - a_1 \cdot 3 + 3a_2x^3 - 5a_2x^2 - 4a_2x - 9a_2 \\ &= (a_1 + 3a_2)x^3 + (-2a_1 - 5a_2)x^2 + (-5a_1 - 4a_2)x + (-3a_1 - 9a_2) \end{aligned}$$

$$\Rightarrow a_1 + 3a_2 = 3$$

$$+2a_1 + 5a_2 = +2$$

$$-5a_1 - 4a_2 = 7$$

$$-3a_1 - 9a_2 = 8$$

Step 1:-

$$\left[\begin{array}{cc|c} 1 & 3 & 3 \\ +2 & 5 & 2 \\ +5 & +4 & -7 \\ +3 & 9 & -8 \end{array} \right] \begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 5R_1 \\ R_4 \rightarrow R_4 - 3R_1 \end{array}$$

$$\left[\begin{array}{cc|c} 1 & 3 & 3 \\ 0 & -1 & -4 \\ 0 & -11 & -22 \\ 0 & 0 & -17 \end{array} \right]$$

Look at the ~~last~~ last row, It's not a completely zero row, It have non-zero in B

$\therefore$ It is Inconsistent

Span Problem! Determine whether the given vectors in

each case is in the Span of S.

HW (a) $(2, -1, 1)$, $S = \{(1, 0, 2), (-1, 1, 1)\}$ $a_1 = 1, b_1 = -1$

HW (b) $(-1, 1, 1, 2)$, $S = \{(1, 0, 1, -1), (0, 1, 1, 1)\}$ Inconsistent

HW (c) $2x^3 - x^2 + x + 3$, $S = \{x^3 + x^2 + x + 1, x^2 + x + 1, x + 1\}$ $\begin{bmatrix} a_1 = -1 \\ a_2 = 3 \\ a_3 = 1 \end{bmatrix}$

C.W (d) $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, $S = \left\{ \begin{pmatrix} 1 & 0 \\ -1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix} \right\}$

(d) Solution

If scalar a_1, a_2, a_3 s.t

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = a_1 \begin{pmatrix} 1 & 0 \\ -1 & 0 \end{pmatrix} + a_2 \begin{pmatrix} 0 & 1 \\ 0 & 1 \end{pmatrix} + a_3 \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} a_1 & 0 \\ -a_1 & 0 \end{pmatrix} + \begin{pmatrix} 0 & a_2 \\ 0 & a_2 \end{pmatrix} + \begin{pmatrix} a_3 & a_3 \\ 0 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} a_1 + a_3 & a_2 + a_3 \\ -a_1 & a_2 \end{pmatrix}$$

$$\Rightarrow a_1 + a_3 = 1, a_2 + a_3 = 0, a_1 = 0, a_2 = 1$$

$$a_1 + a_3 = +1$$

$$a_2 + a_3 = 0$$

$$-a_1 = 0$$

$$a_2 = 1$$

Step 1

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ -1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} +1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

A X B

Step 2

$$[A|B] = \left[\begin{array}{ccc|c} 1 & 0 & 1 & +1 \\ 0 & 1 & 1 & 0 \\ -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 \end{array} \right] R_3 \rightarrow R_3 + R_1$$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 1 & +1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & +1 \\ 0 & 1 & 0 & 1 \end{array} \right] R_4 \rightarrow R_4 - R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 1 & +1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & +1 \\ 0 & 0 & -1 & 1 \end{array} \right] R_4 \rightarrow R_4 + R_3$$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 1 & -1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 2 \end{array} \right] R_3 \rightarrow R_3 - R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 1 & -1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 2 \end{array} \right]$$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 1 & -1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 2 \end{array} \right]$$

Look at the last row it is completely zero. Now.

$\therefore$ Inconsistent $\therefore \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ does not belong to $\text{span}(S)$

Vector space generated by set S:-

A subset S of a vector space V generates $(\text{span}) V$

if $\text{span}(S) = V$, [vector of S , $\text{span } V$]

We call S as generating set. (or) spanning set of V

Example:-

The vectors $(1, 0, 0)$, $(0, 1, 0)$ and $(0, 0, 1)$ generate $\mathbb{R}^3$.

Since an arbitrary vector $(a_1, a_2, a_3) \in \mathbb{R}^3$ is linear combination of these three vectors.

$$\text{(i.e.) } (a_1, a_2, a_3) = a_1(1, 0, 0) + a_2(0, 1, 0) + a_3(0, 0, 1)$$

Example:- In $V = \mathbb{F}^n$, let e_j denote the vector whose j^{th} co-ordinate is 1 & whose other co-ordinates are zero(0).

$$\text{(i.e.) } e_1 = (1, 0, \dots, 0)$$

$$e_2 = (0, 1, 0, \dots, 0)$$

$$e_3 = (0, 0, 1, 0, \dots, 0)$$

$$\vdots$$

$$e_n = (0, 0, \dots, 1)$$

Let $(u_1, u_2, \dots, u_n) \in \mathbb{F}^n$ be an arbitrary element

can be expressed as linear combination of $e_1, e_2, \dots, e_n$.

$$\begin{aligned}(u_1, u_2, \dots, u_n) &= u_1(1, 0, \dots, 0) + u_2(0, 1, \dots, 0) + u_3(0, 0, 1, \dots, 0) \\ &\quad \dots + u_n(0, 0, \dots, 1) \\ &= u_1 e_1 + u_2 e_2 + u_3 e_3 + \dots + u_n e_n\end{aligned}$$

$$\Rightarrow \{e_1, e_2, \dots, e_n\} \text{ generates } \mathbb{F}^n$$

(span)

Problem 1

c.w (1) show that $\mathbb{P}_n(\mathbb{F})$ is generated by $\{1, x, x^2, \dots, x^n\}$.

H.W (2) show that matrices $\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ $\begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ $\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$ generate $M_{2 \times 2}(\mathbb{F})$. (or $M_{2 \times 2}(\mathbb{R})$)

c.w (3) show that vectors $(1, 1, 0)$ $(1, 0, 1)$ $(0, 1, 1)$ generate $\mathbb{F}^3$

Choose an arbitrary element $(a_1, a_2, a_3) \in \mathbb{F}^3$ try to prove

it can be expressed as linear combination of $(1, 1, 0)$ $(1, 0, 1)$ $(0, 1, 1)$

$$\begin{aligned}(a_1, a_2, a_3) &= a_1(1, 1, 0) + a_2(1, 0, 1) + a_3(0, 1, 1) \\ &= (a_1, a_1, 0) + (a_2, 0, a_2) + (0, a_3, a_3) \\ &= (a_1 + a_2, a_1 + a_3, a_2 + a_3)\end{aligned}$$

$$\Rightarrow a_1 + a_2 = a_1$$

$$a_1 + a_3 = a_2$$

$$a_2 + a_3 = a_3$$

$$a_2 = 0$$

$$a_1 - a_2 + a_3 = 0$$

$$a_2 = 0$$

$$a_2 = 0$$

$$\Rightarrow a_1 + a_3 = 0$$

$$a_1 = -1, a_3 = 1$$

Step 1

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$$\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

$(2, 3, 4) \in \mathbb{F}^3$ be a arbitrary element

$$(2, 3, 4) = a_1(1, 1, 0) + a_2(1, 0, 1) + a_3(0, 1, 1)$$

$$= (a_1, a_1, 0) + (a_2, 0, a_2) + (0, a_3, a_3)$$

$$= (a_1 + a_2, a_1 + a_3, a_2 + a_3)$$

$$a_1 + a_2 = 2$$

$$a_1 + a_3 = 3$$

$$a_2 + a_3 = 4$$

$$\Rightarrow \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$$

$$[A|B] = \left[\begin{array}{ccc|c} 1 & 1 & 0 & 2 \\ 1 & 0 & 1 & 3 \\ 0 & 1 & 1 & 4 \end{array} \right] R_2 \rightarrow R_2 - R_1$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 0 & 2 \\ 0 & -1 & 1 & 1 \\ 0 & 1 & 1 & 4 \end{array} \right] R_2 \rightarrow \frac{R_2}{-1}$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 0 & 2 \\ 0 & 1 & -1 & -1 \\ 0 & 1 & 1 & 4 \end{array} \right] R_3 \rightarrow R_3 - R_2$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 0 & 2 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 2 & 5 \end{array} \right]$$

$$a_1 + a_2 = 2$$

$$a_2 - a_3 = -1$$

$$2a_3 = 5$$

$$\boxed{a_3 = 5/2}$$

$$\Rightarrow a_2 - 5 = -1$$

$$\Rightarrow a_2 = -1 + 5 = 4$$

$$\Rightarrow a_1 + 4 = 2 \Rightarrow a_1 = -2$$

$$a_1 = -2, a_2 = 4, a_3 = 5$$

$$\begin{aligned} & -2(1, 1, 0) + 4(1, 0, 1) + 5(0, 1, 1) \\ &= (-2, -2, 0) + (4, 0, 4) + (0, 5, 5) \\ &= (2, 3, 0) \end{aligned}$$

$$a_3 = \frac{5}{2}$$

$$a_2 - \frac{5}{2} = -1$$

$$a_2 = -1 + \frac{5}{2}$$

$$\boxed{a_2 = \frac{3}{2}}$$

$$a_1 + \frac{3}{2} = 2$$

$$a_1 = 2 - \frac{3}{2}$$

$$\boxed{a_1 = \frac{1}{2}}$$

$$\boxed{a_1 = \frac{1}{2} \quad a_2 = \frac{3}{2} \quad a_3 = \frac{5}{2}}$$

Now

$$\begin{aligned} & \frac{1}{2}(1, 1, 0) + \frac{3}{2}(1, 0, 1) + \frac{5}{2}(0, 1, 1) \\ &= \left(\frac{1}{2}, \frac{1}{2}, 0\right) + \left(\frac{3}{2}, 0, \frac{3}{2}\right) + \left(0, \frac{5}{2}, \frac{5}{2}\right) \\ &= (2, 3, 4) \end{aligned}$$

An arbitrary element from $\mathbb{F}^3$ can be written as linear combination of $(1, 1, 0), (1, 0, 1), (0, 1, 1)$

Hence $\{(1, 1, 0), (1, 0, 1), (0, 1, 1)\}$ spans $\mathbb{F}^3 = \mathbb{R}^3 = V$

Linearly Dependent & Linearly Independent

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Initially, while we start Defn of Linear Combination with an example that

$v_1 = (-2, 0, 3)$ can be expressed linear combination of
 $v_2 = (1, 3, 0)$ & $v_3 = (2, 4, -1)$

By the way, we get

$$(-2, 0, 3) = 4(1, 3, 0) + (-3)(2, 4, -1) \quad \text{--- (1)}$$

Hence, we have

$$(2, 4, -1) = \frac{4}{3}(1, 3, 0) + \left(-\frac{1}{3}\right)(-2, 0, 3)$$

or

$$(1, 3, 0) = \frac{1}{4}(-2, 0, 3) + \frac{3}{4}(2, 4, -1)$$

Thus, From the set $\{(-2, 0, 3), (1, 3, 0), (2, 4, -1)\} = S$

one can be written as linear combination of others.

(ie, one vector is depends on others.)

Hence, we say ~~they~~ linearly Dependent Tentative Defn

"In a set one vector is linearly dependent on other."

(ie, S is linearly dependent set.)

otherwise.

$$\textcircled{1} \Rightarrow 4(1, 3, 0) + (-3)(2, 4, -1) + (-1)(-2, 0, 3) = (0, 0, 0) = \vec{0}$$

$$\Rightarrow a_1 v_1 + a_2 v_2 + a_3 v_3 = \vec{0} \text{ where } a_1, a_2, a_3 \in \mathbb{F}$$

Then we make a L.C is Zero with suitable scalar.

Tentative defn

A set $S = \{v_1, v_2, \dots, v_n\}$ of v.s over F is linearly dependent

if $\exists a_1, a_2, \dots, a_n \in F$ s.t

Linear combination of v_i 's must be zero

$$(i.e.) \quad a_1 v_1 + a_2 v_2 + a_3 v_3 + \dots + a_n v_n = 0$$

More Precisely,

$$\textcircled{1} \Rightarrow 4(1, 3, 0) + (-3)(2, 4, -1) + (-1)(-2, 0, 3) = (0, 0, 0) = \vec{0}$$

$$\Rightarrow 4(1, 3, 0) + (-3)(2, 4, -1) + (-1)(-2, 0, 3) + (0)(1, 2, 3) + (0)(2, 3, 4) + (0)(\dots) + \dots + (0)(\dots) = \vec{0}$$

Still the above vectors satisfy the Defn of ~~are~~ linearly dependent.

~~But~~ Basically $\{(1, 3, 0) (2, 4, -1) (-2, 0, 3)\}$ is linearly dependent

Now it's says

$\{(1, 3, 0) (2, 4, -1) (-2, 0, 3) (1, 2, 3) (2, 3, 4) \dots\}$ is linearly dependent

Based on the above Notion, we are going to ③
modify the defn of linearly dependent

Defn of L.D Dependent

A set $S = \{v_1, v_2, \dots, v_n\}$ of v.s over F is said to be

linearly dependent if $\exists a_1, a_2, \dots, a_n \in F$ s.t

Linear combination of vectors from S is zero with ^{some zero scalars} (or) not all scalars zero.

$$(i.e.) \quad a_1 v_1 + a_2 v_2 + \dots + a_n v_n = 0, \text{ with not all } a_i = 0$$

with some a_i 's = 0, but not all

Another example

$(3, 4, 1)$ is not L.C of $\{(1, -2, 1) (-2, -1, 1)\}$

$\therefore (3, 4, 1)$ is not L.D on others.

$\therefore \{(3, 4, 1) \text{ and } (1, -2, 1) (-2, -1, 1)\}$ is not L.D set

Properties of Linearly dependent

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Let $S = \{v_1, v_2, \dots, v_n\}$ of V.s V over F . Then

- ① S is Linearly dependent iff one of the v_i 's is a linear combination of the others
(only ②)
- ② Any superset of L.D set is L.D set. (only ②)
A set contains L.D set is L.D set
- ③ Any set that includes the 0 vector is Linearly dependent.
(Explanation of ③ is later) (After Defining L-Independent)

Linearly Independent:-

Any subset S of V.s V over F is not Linearly dependent, then it is Linearly Independent

For instance, & From H.W

$(3, 4, 1)$ is not linear combination of $\{(1, -2, 1), (-2, -1, 1)\}$
i.e. $\{ (3, 4, 1), (1, -2, 1), (-2, -1, 1) \}$ is not L.D set
(i.e. one cannot be L.C of others.)

$$(3, 4, 1) \neq a_1(1, -2, 1) + a_2(-2, -1, 1)$$

$$\Rightarrow a_1(1, -2, 1) + a_2(-2, -1, 1) + (-1)(3, 4, 1) \neq (0, 0, 0) = \vec{0}$$

i.e. It is not possible to find a_1, a_2, a_3 to make L.C $\vec{0}$.

∴ The only possible way to make the above linear combination Zero is ~~setting~~ all

$$\text{Selecting } a_1 = 0, a_2 = 0, a_3 = 0$$

$$\text{(i.e. } 0(1, -2, 1) + 0(-2, -1, 1) + 0(3, 4, 1) = (0, 0, 0)$$

Based on the above Example, Refine the Defn of L.I.

Def of L.I :-

A set $S = \{v_1, v_2, \dots, v_n\} \subset V$ is said to be L.I if

if L.C of vectors from S is zero with all zero scalar
(The only possible way to make L.C zero is, ^(making) select scalar as zero)

$$(i) \quad a_1 v_1 + a_2 v_2 + \dots + a_n v_n = 0, \Rightarrow a_i = 0 \quad 1 \leq i \leq n.$$

Properties of L.I :-

Let $S = \{v_1, v_2, \dots, v_n\} \subset V$, Then

① S is L.I iff Any vector v_i is not a L.C of others.

② Any subset of L.I set is L.I set

—————x—————x—————
Method to check L.I / L.D set

Step 1 Arrange the given vectors as rows in a matrix
and perform row operations until the row
echelon form is reached

Step 2 Declare the result based on Following

(or) (i) If there is a zero row in echelon form,
Then the set of vectors is L.D

(ii) If there is no zero row in echelon form,
Then the set of vectors is L.I

Problem 1: Determine if the following sets are linearly depd (or) Indepd

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Hue (a) $\{(1, 2, 0), (0, 3, 1), (-1, 0, 1)\}$ in $\mathbb{R}^3$ (Indepd)

Hue (b) $\{(1, -1, 2), (1, -2, 1), (1, 1, 4)\}$ in $\mathbb{R}^3$ (depd)

(C.W) (c) $\{(1, 1, 2, 4), (2, -1, -5, 2), (1, -1, -4, 0), (2, 1, 1, 6)\}$ in $\mathbb{R}^4$ (depd)

Hue (d) $\{(1, 1, 0, 0), (0, 1, -1, 0), (0, 0, 0, 3)\}$ in $\mathbb{R}^4$ (Indepd)

(C.W) (e) $\{x^3 + 2x^2, -x^2 + 3x + 1, x^3 - x^2 + 2x - 1\}$ in $\mathbb{R}^3$ (Indepd)

Hue (f) $\left\{ \begin{pmatrix} 1 & -3 \\ -2 & 4 \end{pmatrix}, \begin{pmatrix} -2 & 6 \\ 4 & -8 \end{pmatrix} \right\}$ in $M_{2 \times 2}(\mathbb{R})$ (Depd)

C.W (g) $\left\{ \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \begin{pmatrix} 2 & -1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 3 & -2 \\ 1 & 2 \end{pmatrix} \right\}$ in $M_{2 \times 2}(\mathbb{F})$ (Indepd)

Sol:

(c) Consider the matrix of vectors

$$A = \begin{bmatrix} 1 & 1 & 2 & 4 \\ 2 & -1 & -5 & 2 \\ 1 & -1 & -4 & 0 \\ 2 & 1 & 1 & 6 \end{bmatrix} \begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - R_1 \\ R_4 \rightarrow R_4 - 2R_1 \end{array}$$

$$\sim \begin{bmatrix} 1 & 1 & 2 & 4 \\ 0 & -3 & -9 & -6 \\ 0 & -2 & -6 & -4 \\ 0 & -1 & -3 & -2 \end{bmatrix} \begin{array}{l} R_2 \rightarrow \frac{R_2}{-3} \\ R_3 \rightarrow \frac{R_3}{-2} \\ R_4 \rightarrow -R_4 \end{array}$$

$$\sim \begin{bmatrix} 1 & 1 & 2 & 4 \\ 0 & 1 & 3 & 2 \\ 0 & 1 & 3 & 2 \\ 0 & 1 & 3 & 2 \end{bmatrix} \begin{array}{l} R_3 \rightarrow R_3 - R_2 \\ R_4 \rightarrow R_4 - R_2 \end{array}$$

$$\sim \begin{bmatrix} 1 & 1 & 2 & 4 \\ 0 & 1 & 3 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

It has Zero rows,

$\therefore$ Given vectors are Linearly dep^t.

———— x ——— x ———

(e) Consider the matrix of vectors

$$A = \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & -1 & 3 & 1 \\ 1 & -1 & 2 & -1 \end{bmatrix} \quad R_3 \rightarrow R_3 - R_1$$

$$\sim \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & -1 & 3 & 1 \\ 0 & -3 & 2 & -1 \end{bmatrix} \quad \begin{array}{l} R_2 \rightarrow -R_2 \\ R_3 \rightarrow -R_3 \end{array}$$

$$\sim \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 1 & -3 & -1 \\ 0 & 3 & -2 & 1 \end{bmatrix} \quad \begin{array}{l} R_1 \\ R_3 \rightarrow R_3 - 3R_2 \end{array} \quad \begin{array}{l} -2+9 \\ 1+3 \end{array}$$

$$\sim \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 1 & -3 & -1 \\ 0 & 0 & 7 & 4 \end{bmatrix}$$

$\sim$ Echelon Form

~~an~~ There is no Zero row in Echelon form,

$\therefore$ Given vectors are Linearly Independent

———— x ——— x ———

(8) Consider the vector matrix of vectors

$$A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 2 & -1 & 1 & 0 \\ 3 & -2 & 1 & 2 \end{bmatrix} \quad \begin{array}{l} R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 3R_1 \end{array} \quad \begin{array}{l} 2-12 \\ 3-12 \end{array}$$

$$\sim \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & -5 & -5 & -8 \\ 0 & -8 & -8 & -10 \end{bmatrix} \quad \begin{array}{l} R_2 \rightarrow \frac{R_2}{-5} \\ R_3 \rightarrow -\frac{R_3}{2} \end{array}$$

$$\sim \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 1 & 8/5 \\ 0 & 4 & 4 & 5 \end{bmatrix} \quad R_3 \rightarrow R_3 - 4R_2$$

$$\sim \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 1 & 8/5 \\ 0 & 0 & 0 & -7/5 \end{bmatrix} \quad \begin{array}{l} 5 - 4 \cdot \frac{8}{5} \\ \frac{25 - 32}{5} \end{array}$$

$\sim$ Echelon Form.

There is no zero row, in Echelon Form

$\therefore$ Given Vectors are L.I.

————— $\times$ ————— $\times$ —————

Problem

- ① S.T the set $\{1, x, x^2, \dots, x^n\}$ is linearly independent in $P_n(F)$
- ② S.T the set $\{e_1, e_2, \dots, e_n\} \subset F^n$ is linearly independent in F^n

Bases & Dimension!

Basis! of V

A subset S of a vector space V over F is said to be basis of $V(F)$ if

(i) S consists of linearly independent vectors.

(ii) S generates (spans) V

(ie, $V = \text{span}(S)$)

(ie each vector in V is a linear combination of vectors from S .)

Dimension! of a vector space

If a set B is a Basis set of $V(F)$, then

No of vectors in B is called dimension of V

(ie, $\dim(V) = \text{No of vectors in Basis } B$.)

— x — x —

Example!.

① A set S consisting of n vectors $\{e_1, e_2, \dots, e_n\} = B$

where $e_i = (0, 0, \dots, \underset{i\text{th}}{1}, \dots, 0)$ $1 \leq i \leq n$.

is a Basis set of $\mathbb{R}^n$

Since $\{e_1, e_2, \dots, e_n\}$ is linearly independent

& an arbitrary element $v = (a_1, a_2, \dots, a_n) \in \mathbb{R}^n$, can be expressed as linear combination of $e_1, e_2, \dots, e_n$.

This B is called std basis for $\mathbb{R}^n$.

— x — x —

11/4

- (2) $B = \{(1,0), (0,1)\}$ is a basis for $\mathbb{R}^2$
- (3) $B = \{(1,0,0), (0,1,0), (0,0,1)\}$ is also basis of $\mathbb{R}^3$.
- (4) $B = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \right\}$ is std basis for $M_{2 \times 2}(\mathbb{F})$
- (5) The set $\{1, x, x^2, \dots, x^n\}$ is basis for $P_n(\mathbb{F})$. This basis is called standard basis for $P_n(\mathbb{F})$.

(6) Basis set for Zero space $\{0\}$

$\emptyset = B = \{ \}$ is linearly independent.

$\emptyset = B = \{ \}$ spans $\{0\}$

$\therefore B = \emptyset$ is a Basis for Zero space $\{0\}$

— x — x —

Theorem:-

Let V be a Vector space and $B = \{u_1, u_2, \dots, u_n\}$ be a subset of V . Then B is a ~~set~~ basis for V

iff

$\forall v \in V$ can be uniquely expressed as a linear combination of vectors of B ,

i.e., v can be expressed in the form

$$v = a_1 u_1 + a_2 u_2 + \dots + a_n u_n, \text{ for unique scalars.}$$

Proof: Part 1:-

Given that, V is a Vector space over $\mathbb{F}$

$\< B = \{u_1, u_2, \dots, u_n\}$ is a Basis for V — (1)

To P.T:- $\forall v \in V$, can be uniquely expressed as L.C of vectors of B

i.e., $v = a_1 u_1 + a_2 u_2 + \dots + a_n u_n$, $(a_1, a_2, \dots, a_n)$ are unique scalars

Proof:

As B is a Basis of V , $\Rightarrow V = \text{span}(B)$

$\Rightarrow \forall v \in V$ s.t.

$$v = a_1 u_1 + \dots + a_n u_n, \text{ for } a_i \in \mathbb{F}, u_i \in B, 1 \leq i \leq n.$$

②

It is enough to show that Expression ② is unique.

For: Assume the contrary,

$\exists b_1, b_2, \dots, b_n \in \mathbb{F}$ s.t.

$$v = b_1 u_1 + b_2 u_2 + \dots + b_n u_n \quad \text{--- ③}$$

$$\text{②} \text{ \& } \text{③} \Rightarrow a_1 u_1 + \dots + a_n u_n = b_1 u_1 + \dots + b_n u_n$$

$$\Rightarrow (a_1 - b_1) u_1 + \dots + (a_n - b_n) u_n = 0$$

$\Rightarrow$ L.C of vectors from B is Zero.

$\Rightarrow$ L.C of linearly independent vectors from B is Zero

$\Rightarrow$ The only possible way to make the L.C of vectors is Zero, for linear independent vectors,

$$\Rightarrow \forall a_i - b_i = 0 \quad 1 \leq i \leq n$$

$$\Rightarrow a_i = b_i$$

^{Expressions}
 $\Rightarrow$ ② & ③ are same.

Thus $v \in V$, can be expressed uniquely as L.C of vectors of B .

Part II:

Given that, V is vector space over F
 Each vector $v \in V$ expressed uniquely as L.C of
 vectors of B

$$(i) v = a_1 u_1 + \dots + a_n u_n, a_i \in F, u_i \in B. \quad \text{--- (1)}$$

To P.T : B is Basis

Proof: For that, we need to prove

(ie) (i) $B = \{u_1, u_2, \dots, u_n\}$ is linearly independent vectors

$$(ii) V = \text{span}(B)$$

Proof:

For (i) - As $0 \in V$, From (1)

$$0 = b_1 u_1 + b_2 u_2 + \dots + b_n u_n \quad b_i \in F, u_i \in B \quad \text{--- (2)}$$

$$\text{But } 0 = 0u_1 + 0u_2 + \dots + 0u_n$$

By uniqueness of (1)

$$b_1 = b_2 = \dots = b_n = 0 \quad \text{--- (3)}$$

From (2) & (3), ~~b_i~~

$\{u_1, u_2, \dots, u_n\}$ is linearly independent

For (ii)

The given statement Prove, that

B spans (generates) V

$$(i) V = \text{span}(B)$$

Hence the Theorem

Property:-

Let V be a vector space with dimension ' n '.

(a) Any L.I subset of V that contains exactly ' n ' vectors is a basis for V .

(b) Every L.I subset of V can be extended to a basis of V .

Problem..

The Vectors $\{(2, -3, 5), (8, -12, 20), (1, 0, -2), (0, 2, -1), (7, 2, 0)\}$ generates $\mathbb{R}^3$.

Find a subset of the given set that is a basis for $\mathbb{R}^3$.

Soln. -

Given that

$$S = \{(2, -3, 5), (8, -12, 20), (1, 0, -2), (0, 2, -1), (7, 2, 0)\}.$$

We need Construct Basis $B \subset S$ of $\mathbb{R}^3$

~~Step~~ W.K.T $\dim \mathbb{R}^3 = 3 \Rightarrow B$ should contains 3 Vectors.
& it should be L.I

Let us begin with

Step 1

$$B = \{(2, -3, 5)\}$$

Step 2 Argue with $(8, -12, 20)$ Check L.I with $(8, -12, 20)$ of B

obviously, $(8, -12, 20) = 4(2, -3, 5)$

$\therefore (8, -12, 20)$ L.D on $(2, -3, 5)$, we should exclude
 $(8, -12, 20)$ from B .

Step 3.. Argue with $(1, 0, -2)$
check L.I of B with $(1, 0, -2)$

$$\begin{bmatrix} 2 & -3 & 5 \\ 1 & 0 & -2 \end{bmatrix} R \leftarrow 2R_2$$

$$\sim \begin{bmatrix} 1 & 0 & -2 \\ 2 & -3 & 5 \end{bmatrix} R_2 \rightarrow R_2 - 2R_1$$

$$\sim \begin{bmatrix} 1 & 0 & -2 \\ 0 & -3 & 9 \end{bmatrix} \sim \text{Echelon form}$$

does not have a complete zero row

$\therefore (2, -3, 5), (1, 0, -2)$ form L.I set
Hence $B = \{(2, -3, 5), (1, 0, -2)\}$

Step 4:-

(4)

Check B with $(0, 2, -1)$

$$\begin{bmatrix} 2 & -3 & 5 \\ 1 & 0 & -2 \\ 0 & 2 & -1 \end{bmatrix} \quad R_1 \leftrightarrow R_2$$

$$\sim \begin{bmatrix} 1 & 0 & -2 \\ 2 & -3 & 5 \\ 0 & 2 & -1 \end{bmatrix} \quad R_2 \rightarrow R_2 - 2R_1$$

$$\sim \begin{bmatrix} 1 & 0 & -2 \\ 0 & -3 & 9 \\ 0 & 2 & -1 \end{bmatrix} \quad R_2 \rightarrow \frac{R_2}{-3}$$

$$\sim \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & -3 \\ 0 & 2 & -1 \end{bmatrix} \quad R_3 \rightarrow R_3 - 2R_2$$

$$\sim \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & -3 \\ 0 & 0 & 5 \end{bmatrix}$$

There is no Zero Row in Echelon form.

Hence B includes $(0, 2, -1)$

$$B = \{ (2, -3, 5) (1, 0, -2) (0, 2, -1) \}$$

Step 5

No need to check B with last vector $(7, 2, 0)$

Since Basis of $\mathbb{R}^3$ should contain 3-linearly independent vectors

$$\text{Thus } B = \{ (2, -3, 5) (1, 0, -2) (0, 2, -1) \} \text{ Form Basis of } \mathbb{R}^3$$

— x —

Result:

Let W be a Subspace of a Finite dimension V .
Then W is Finite dimension.

$$\dim(W) \leq \dim V.$$

Also if $\dim W = \dim V$ Then $V = W$.

Problem

① Determine which of the following sets are bases

for $\mathbb{R}^3$

(a) $\{(1, 0, -1), (2, 5, 1), (0, -4, 3)\}$ L.I, Form a basis

(b) $\{(2, -4, 1), (0, 3, -1), (6, 0, -1)\}$ L.D not a basis

(c) $\{(1, -3, -2), (-3, 1, 3), (-2, 10, -2)\}$ L.D, not basis

② Determine which of the following sets are bases for

$\mathbb{P}_2(\mathbb{R})$

(a) $\{-1-x+2x^2, 2+x-2x^2, 1-2x+4x^2\}$ L.D

(b) $\{1-2x-2x^2, -2+3x-x^2, 1-x+6x^2\}$ L.I

Result

If W_1 & W_2 are Finite dimensional Subspace of a Vector space V , Then the Subspace W_1+W_2 is finite dim and

$$\dim(W_1+W_2) = \dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2)$$